



AI-Powered Data Analytics Internship

Week 2 Submission:

Cohort Analysis Report with visual insights

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Executive Summary

This report presents the Week 2 deliverables building on the cleaned learner participation dataset established in Week 1. Working from the processed dataset of 8,558 records across 34 features, the team carried out data standardization and feature engineering, exploratory data analysis, quarterly trend analysis, opportunity category analysis, trend visualizations, correlation and heatmap analysis, and the identification of forecasting signals to support predictive modelling in the following phase. Across these analyses, Internship opportunities were confirmed as the dominant category by volume (63.3% of applications) but also the most competitive, with a participation rate far below Course, Event, Engagement, and Competition opportunities. Engineered features such as `total_category_opportunities` and `participation_frequency` emerged as the strongest behavioural signals for future modelling, while quarterly and seasonal patterns highlighted clear peaks in learner activity. An AI-assisted analysis phase was also carried out to surface non-obvious patterns and validate findings against the underlying dataset. Together, these deliverables give the team a well-documented, evidence-based foundation for Week 3 predictive modelling.

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1. Introduction and Objectives

This project focuses on understanding and predicting learner participation in RIT learning opportunities using the historical signup and application dataset from the platform. Week 1 established a cleaned dataset, a data dictionary, and a summary statistics report. Week 2 builds directly on that foundation, with the objective of preparing the dataset for predictive modelling through deeper analysis. The specific objectives for this phase were to standardize and engineer meaningful features from the cleaned data, explore the dataset through descriptive statistics and visualizations, analyse historical and quarterly participation trends, compare performance across opportunity categories, and identify the strongest forecasting signals for participation outcomes. Each of the following sections documents one part of this work, culminating in a set of features and insights that will directly inform the Week 3 modelling phase.

2. Data Standardization & Feature Engineering

Data Standardization

To improve consistency, the column names were standardized by converting all text to lowercase, removing unnecessary spaces, and replacing spaces and forward slashes with underscores. This naming convention makes the dataset easier to work with and reduces errors during analysis. Categorical variables were also standardized by removing extra spaces and converting text values into Title Case. This ensured that categories such as gender, country, institution name, and opportunity category were recorded consistently throughout the dataset.

Date Transformation

Several date columns were converted into the datetime format to enable time-based analysis. These included the learner's date of birth, signup date, application date, opportunity start date, and opportunity end date. Once the dates were converted, additional temporal features such as application year, application month, application quarter, signup year, signup month, and signup quarter were extracted. These variables support the analysis of participation trends and learner cohorts over time.

Feature Engineering

Several new variables were created to enrich the dataset and provide deeper insights into learner participation.

Engagement Duration

The `engagement_duration_days` feature was created by calculating the difference between the `opportunity_end_date` and `opportunity_start_date` for each program. This variable represents the total duration of an opportunity in days, indicating the length of time learners could participate. The initial results showed a duration of 604 days for the first five records, suggesting that these learners participated in the same program or in programs with identical durations.

The `engagement_duration_days` feature was engineered to quantify the length of each learning opportunity, enabling analysis of program duration alongside learner participation and engagement. This variable supports further investigation into whether the length of a program influences participation rates, learner retention, or completion outcomes, making it a valuable predictor for understanding engagement patterns across different opportunities.

Participation Frequency and Repeat Participant

Participation Frequency and Repeat Participant Indicator: The `participation_frequency` feature was created by counting the total number of opportunities each learner participated in, using `first_name` and `institution_name` to group individual learners. Based on this count, a binary variable, `is_repeat_participant`, was generated, where 0 indicates a learner who participated only once and 1 indicates a learner who participated in more than one opportunity. The initial results showed a mix of first-time and repeat participants, demonstrating varying levels of learner engagement across the platform.

The `participation_frequency` and `is_repeat_participant` features were engineered to measure learner retention and recurring engagement. These variables provide valuable insights into participation behaviour by identifying learners who consistently return for additional opportunities. They also support further analysis of engagement patterns and

can serve as important predictors in understanding learner loyalty, participation trends, and long-term involvement with the platform.

Total Category Opportunities

The `total_category_opportunities` feature was created by grouping the dataset based on `opportunity_category` and calculating the number of unique `opportunity_ids` available within each category. The resulting count was assigned to every record belonging to the respective category. The analysis showed that Internship had the highest number of unique opportunities (9), followed by Competition (5), Course (4), Event (3), and Engagement (2). For example, all records under the Course category were assigned a value of 4, indicating that there were four unique course opportunities available.

This feature was engineered to measure the diversity and availability of opportunities across different categories. It helps identify which categories provide more opportunities and supports comparisons of learner participation patterns across categories. The feature can also be useful in understanding whether the availability of opportunities within a category influences learner engagement and participation trends.

Days to Apply (Onboarding Conversion Time)

The `days_to_apply` feature was created to measure the time taken by learners to submit an application after registering on the platform. This was calculated by finding the difference between the `apply_date` and `learner_signup_datetime`. The resulting values were rounded up to whole numbers, while negative values caused by inconsistent date records were replaced with 0. Missing application dates were retained as missing values (NaN). The output showed that some learners applied within one day of registration, while others took longer periods, such as 42 days, indicating differences in onboarding engagement speed.

This feature was engineered to evaluate learner activation and understand onboarding conversion behaviour. It helps identify how quickly new users move from registration to participation, allowing analysis of engagement patterns, user responsiveness, and potential barriers in the onboarding process. The feature can also support predictive

analysis by helping determine factors that influence how soon learners become active on the platform.

Onboarding Speed Binning

The `apply_speed_category` feature was created by grouping learners into different categories based on the number of days taken to apply after signing up on the platform (`days_to_apply`). Learners were classified into six groups: Same Day (<1d), Within 1 Week, within 1 Month, 1–3 Months, 3+ Months, and Not Recorded for missing application dates. This categorization helps summarize onboarding behaviour by showing how quickly learners transition from registration to active participation. The results showed that some learners applied immediately after registration, while others required weeks or months before engaging with opportunities.

This feature was engineered to simplify the analysis of onboarding patterns and improve the understanding of learner engagement behavior. By converting continuous time values into meaningful groups, it becomes easier to identify trends, compare user activation speeds, and determine whether learners who apply quickly have different participation patterns compared to those who take longer to engage. This feature can also support segmentation and predictive modelling of learner conversion behavior.

Duration Binning

The `duration_category` feature was created by grouping opportunities based on their calculated duration in days (`engagement_duration_days`). Opportunities were classified into six categories: Short (<1 mo), Medium (1–3 mos), Long (3–6 mos), Year-Long, Multi-Year, and Unknown/Ongoing for opportunities with missing duration information. This categorization provides a clearer view of opportunity length and allows easier comparison between programs. The initial results showed that the first five records were categorized as multi-year, indicating that these opportunities lasted for more than 365 days, consistent with the previously calculated duration of 604 days.

This feature was engineered to transform continuous duration values into meaningful groups that are easier to interpret and analyse. By categorizing opportunities based on

length, the analysis can identify patterns in learner participation across short-term and long-term programs. It also helps evaluate whether opportunity duration influences learner engagement, participation frequency, and overall program performance.

Participant Age

The age feature was created by calculating the estimated age of each learner using their `date_of_birth` and January 1, 2024, as the reference date. The calculation involved finding the difference between the reference date and each learner's date of birth, then converting the result from days into years using 365.25 days to account for leap years. The first five records showed learner ages of 22, 23, 21, 24, and 23 years, indicating that the initial participants were mainly within the early adult age range.

This feature was engineered to understand the demographic characteristics of learners and analyse participation patterns across different age groups. Age information can help identify trends in learner engagement, compare participation behavior among age segments, and support further analysis of how demographic factors may influence platform usage and opportunity participation.

Outcome Features (Status Metrics)

The `is_active_participation` and `is_dropped_or_rejected` features were created to capture learner participation outcomes based on their `status_description`. The `is_active_participation` indicator assigns a value of 1 to learners with statuses such as Started, Team Allocated, and Rewards Award, while all other statuses are assigned 0. Similarly, the `is_dropped_or_rejected` indicator assigns a value of 1 to learners with statuses including Withdraw, Dropped Out, and Rejected, with other statuses assigned 0. These features provide a simplified way to identify learners who actively participated and those who discontinued or were not accepted into opportunities.

These features were engineered to transform application status information into measurable indicators that can be easily analysed. They help evaluate learner engagement outcomes, identify participation success or dropout patterns, and support further analysis of factors influencing learner retention and completion. Together with other engineered

features such as engagement duration, application speed, age, and participation frequency, these metrics provide a comprehensive understanding of learner behavior and platform performance.

SUMMARY OF FEATURE ENGINEERING

Feature Engineered Name	Description	Purpose / Logic Behind It
Engagement Duration (engagement_duration_days)	Calculated as the difference between opportunity_end_date and opportunity_start_date, representing the total duration of each learning opportunity in days	Measures programme length to examine whether opportunity duration influences learner participation, engagement, retention, and completion outcomes.
Participation Frequency (participation_frequency)	Counts the total number of opportunities each learner participated in by grouping records using first_name and institution_name.	Quantifies learner engagement and retention by identifying how frequently learners participate in opportunities, supporting analysis of loyalty and long-term engagement.
Repeat Participant Indicator (is_repeat_participant)	Binary feature derived from participation_frequency, where 0 indicates a first-time participant and 1 indicates a learner who participated in multiple opportunities.	Identifies repeat learners, enabling analysis of recurring engagement, learner retention, and participation behaviour across the platform
Total Category Opportunities (total_category_opportunities)	Counts the number of unique opportunities	Measures the availability and diversity of

	available within each opportunity_category and assigns the corresponding value to all records in that category.	opportunities across categories, allowing analysis of whether opportunity supply influences learner participation patterns.
Days to Apply (days_to_apply)	Calculated as the number of days between learner_signup_datetime and apply_date, with negative values replaced by 0 and missing application dates retained as NaN.	Evaluates onboarding conversion speed by measuring how quickly learners progress from registration to application, supporting analysis of activation and engagement behaviour.
Onboarding Speed Category (apply_speed_category)	Categorises learners into onboarding groups (Same Day, Within 1 Week, Within 1 Month, 1–3 Months, 3+ Months, and Not Recorded) based on days_to_apply.	Simplifies analysis of onboarding behaviour by grouping learners according to application speed, enabling segmentation and comparison of engagement patterns.
Duration Category (duration_category)	Groups opportunities into categories such as Short, Medium, Long, Year-Long, Multi-Year, and Unknown/Ongoing using engagement_duration_days.	Converts continuous duration values into meaningful categories, making programme length easier to interpret and compare during analysis.
Participant Age (age)	Estimated from date_of_birth using 1 January 2024 as the	Captures demographic characteristics to analyse participation trends across

	reference date and converting the age difference into years.	age groups and assess the influence of age on learner engagement.
Active Participation Indicator (is_active_participation)	Binary outcome variable assigning 1 to learners with active participation statuses (e.g., Started, Team Allocated, Rewards Award) and 0 otherwise.	Provides a simplified measure of active learner engagement, supporting analysis of participation outcomes and success rates.
Dropped or Rejected Indicator (is_dropped_or_rejected)	Binary outcome variable assigning 1 to learners with statuses such as Withdraw, Dropped Out, or Rejected and 0 otherwise.	Identifies learners who discontinued or were unsuccessful, enabling investigation of dropout patterns, learner retention, and factors influencing unsuccessful participation.

3. Exploratory Data Analysis (Summary Statistics)

Key Features Analysed

- age: User demographic age.
- days_to_apply: Time taken by users to submit applications.
- engagement_duration_days: Total duration of active user engagement.
- participation_frequency: How often users participate in events or tasks.
- total_category_opportunities: Total available opportunities within user categories.

Descriptive Statistics

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age	8558. 0	23.75	23.0	4.37	3.0	4.0	22. 0	25.0	57.0
days_to_apply	8024. 0	57.65	5.0	89.08	92.0	0.0	1.0	93.0	427. 0
engagement_duration_ days	3473. 0	373.0 4	468.0	290.6 8	541. 0	0.0	63. 0	604. 0	913. 0
participation_frequency	8558. 0	5.76	4.0	5.75	8.0	1.0	1.0	9.0	34.0
total_category_opportu nities	8558. 0	7.12	9.0	2.50	5.0	2.0	4.0	9.0	9.0

User ages span from 4 to 57 years, showing broad overall appeal. However, with a median age of around 24 years and 50 per cent of users falling between 23 and 25 years , the primary target audience driving activity is firmly young adults.

On average, users take *57 days** to submit an application. This multi-week timeframe shows that users do not make impulsive choices; instead, they take significant time to evaluate opportunities before committing.

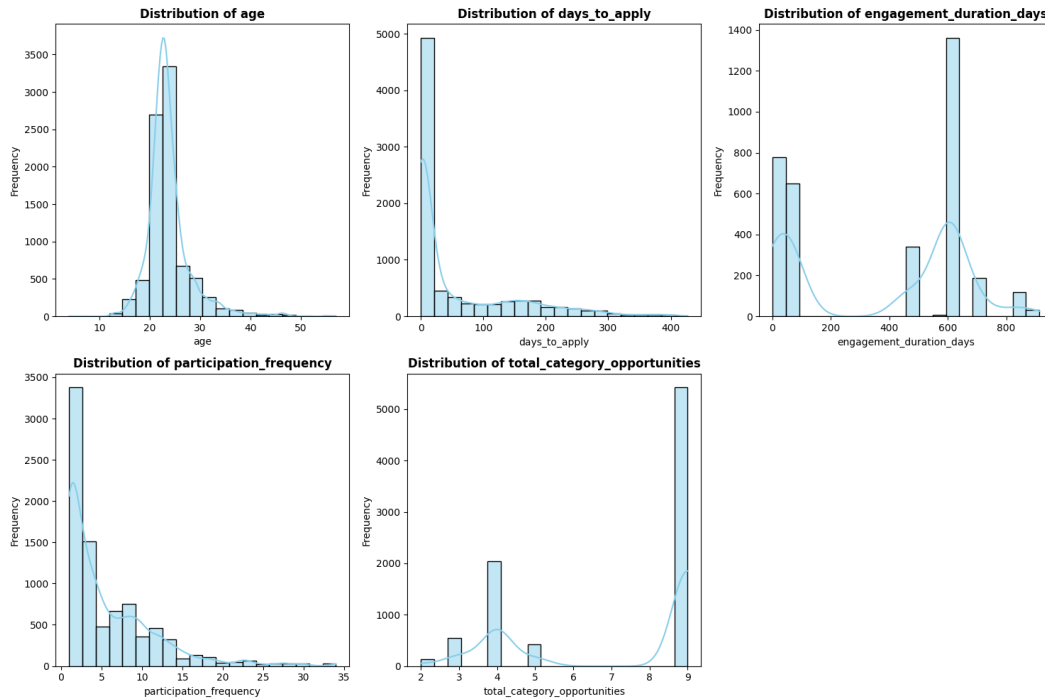
Active engagement ranges widely from **0 to 913 days**. The fact that the median engagement duration is significantly lower than the upper limit reveals a heavy right-skew: while most users stay for a short duration, a small subset of "power users" remains active for nearly 2.5 years (913 days)

The data highlights regular re-engagement across user profiles, showing that active members return consistently rather than visiting the platform just once.

Users engage with a median of 7 total opportunities within their preferred categories. This indicates high platform involvement, as users actively explore multiple options rather than applying to a single post.

While the mean time to apply is 57.65 days, the median is only 5 days. This large gap between mean and median, along with a maximum of 427 days and a standard deviation (std) of 89.08 , shows that most users apply almost immediately within a week, while a small group of late applicants heavily drags up the average

Distribution Analysis



age: The distribution is symmetric and unimodal peaking sharply between 22 and 25 years old, which occurs because the platform primarily attracts students and recent graduates entering the job market.

days_to_apply: The distribution is extremely right-skewed with an L-shape where nearly 5,000 users submit applications within the first 25 days, showing that application decisions are driven by immediate interest upon discovery.

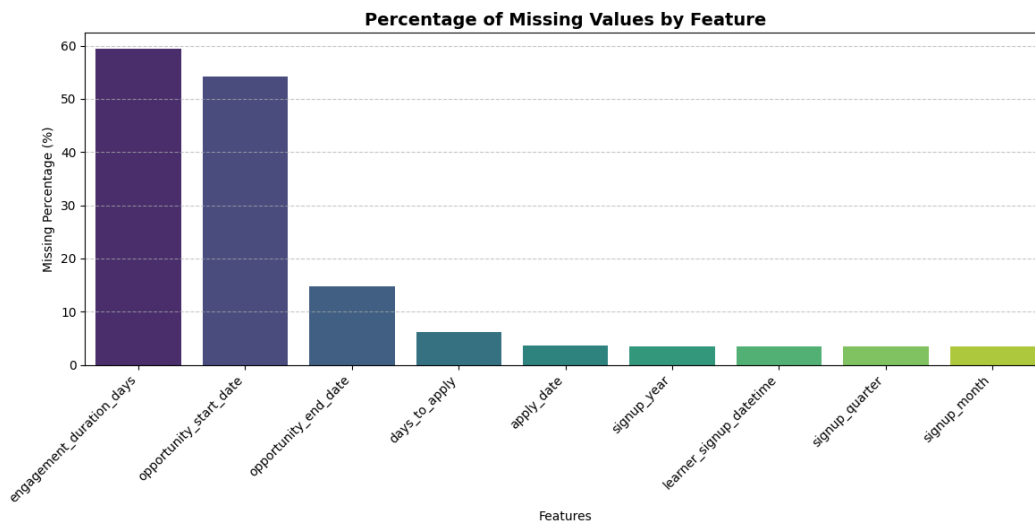
engagement_duration_days: The distribution is bimodal with two distinct peaks under 100 days and around 600 days, reflecting a retention divide where users either churn quickly or become fully embedded.

participation_frequency: The distribution is heavily right-skewed following a power-law curve where over 3,300 users interact only 1 to 2 times, indicating a steep initial trial phase where casual users evaluate the platform before committing to regular activity.

total_category_opportunities: The distribution is discrete and bimodal with clusters at 4 and 9 opportunities, demonstrating that engaged users actively seek out and exhaust all available choices.

Missing Values Summary

Columns	Missing_Count	Missing_Percentage (%)
engagement_duration_days	5085	59.42
opportunity_start_date	4637	54.18
opportunity_end_date	1262	14.75
days_to_apply	534	6.24
apply_date	307	3.59
signup_year	295	3.45
learner_signup_datetime	295	3.45
signup_quarter	295	3.45
signup_month	295	3.45



`engagement\_duration\_days`: Exhibiting the highest missingness rate across the entire dataset at nearly 60%, this high proportion occurs because duration tracking is tied directly to active account qualification and milestone entry. This indicates that missing values are not random system errors, but rather represent unengaged or early drop-off users who left the platform prior to reaching their first measurable milestone.

`opportunity\_start\_date`: Showing a high missing rate above 50%, this pattern reflects a structural data entry pattern where specific opportunity pairings were either unscheduled, open-ended, or left unassigned in the system pipeline at the time of entry.

opportunity_end_date`:Displaying a moderate missingness rate of around 15%, these records represent ongoing or rolling opportunities that operate continuously without a fixed closing date.

`days\_to\_apply`:Containing a low missingness rate around 6%, these unpopulated records correspond directly to users who registered profiles and actively viewed listings on the platform, but ultimately chose not to submit a formal application.

`apply\_date`: Featuring a minor missing proportion under 4%, these records represent unsubmitted application entries where profile interaction took place without a finalized submission timestamp.

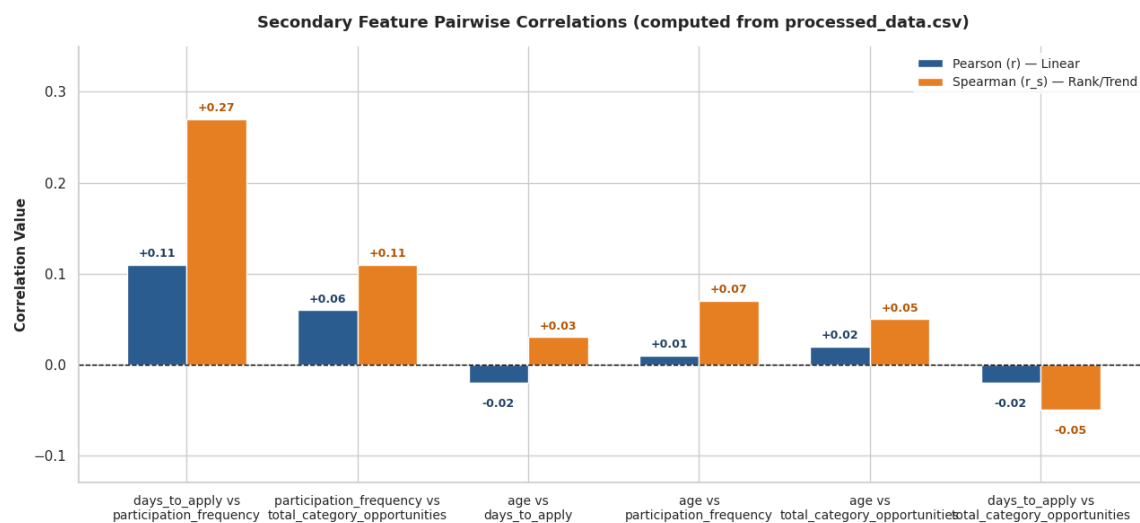
signup_year`, `learner_signup_datetime`, `signup_quarter`, and `signup_month`: Sharing an identical low missing baseline below 4%, this uniform block represents a specific subset of user profile entries that failed to record foundational registration metadata during account creation.

Correlation Analysis

Feature Pair	Pearson Linear (r)	Spearman Rank (r_s)	Relationship Status
days_to_apply ⇌ participation_frequency	+0.11	+0.27	Weak-to-Moderate Monotonic Positive
participation_frequency ⇌ total_category_opportunities	+0.06	+0.11	Negligible Relationship
age ⇌ days_to_apply	-0.02	+0.03	Independent (No Relationship)

age ⇄ participation_frequency	+0.01	+0.07	Independent (No Relationship)
age ⇄ total_category_opportunities	+0.02	+0.05	Independent (No Relationship)
days_to_apply ⇄ total_category_opportunities	-0.02	-0.05	Independent (No Relationship)

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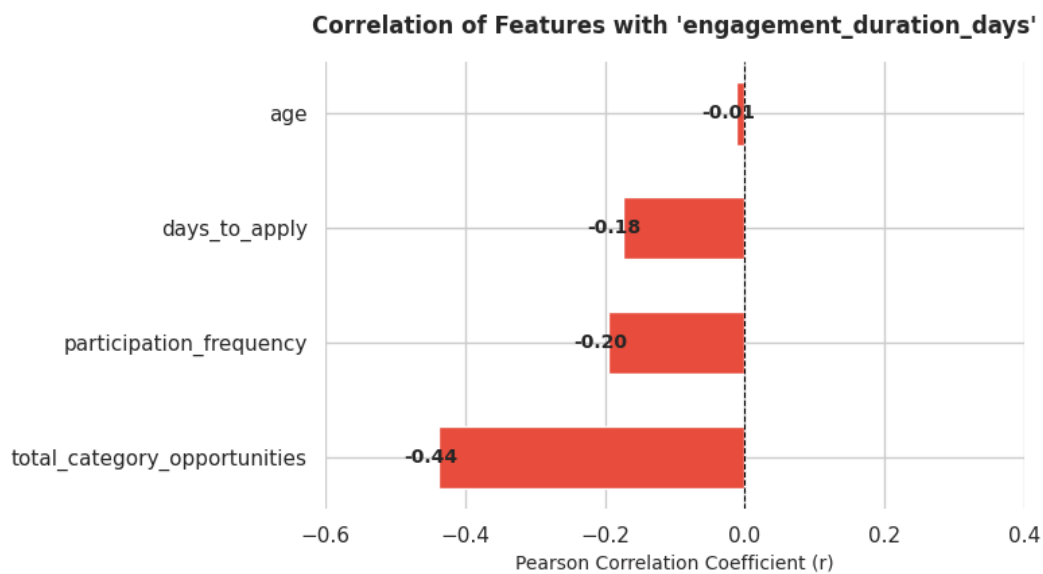


days_to_apply vs participation_frequency: The Spearman rank correlation (+0.27) is notably stronger than the Pearson linear value (+0.11), which reveals a non-linear monotonic trend where users with higher overall platform engagement tend to take slightly longer before submitting their applications.

participation_frequency vs total_category_opportunities: Displays a negligible positive association (+0.06 Pearson, +0.11 Spearman), indicating that the sheer volume of category options available on the platform has little direct bearing on how frequently a user actually participates.

Demographic Independence of age: The variable age shows near zero correlations across all pairs (Pearson and Spearman values ranging between -0.02 and +0.07), demonstrating that user engagement patterns, application speeds, and category exploration are entirely independent of age.

days_to_apply vs total_category_opportunities: Shows virtually zero correlation (-0.02 Pearson, -0.05 Spearman), proving that the number of opportunities presented to a user does not impact or delay their decision timeline when applying



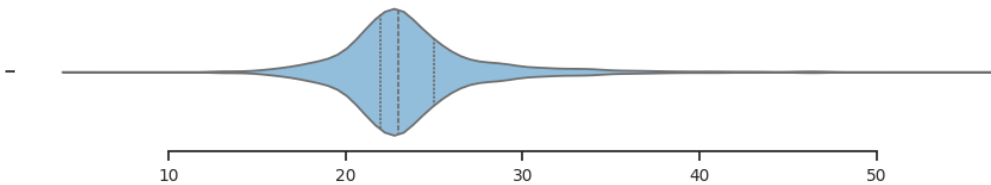
total_category_opportunities`: Shows the strongest relationship with engagement duration ($r = -0.44$). This moderate negative trend suggests that users operating in categories with fewer total opportunities tend to maintain active engagement over longer timeframes.

participation_frequency` and `days_to_apply` Trends: Both features display weak negative correlations ($r = -0.20$ and $r = -0.18$, respectively) with engagement duration, indicating that higher action frequency and slower application speeds correlate slightly with shorter overall active windows.

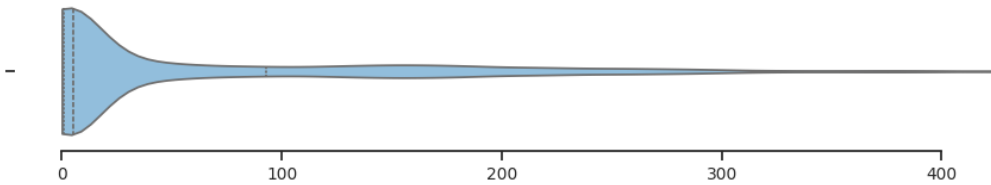
Age Independence: The 'age' feature displays virtually zero correlation ($r = -0.01$), reconfirming that user age plays no role in determining how long a user remains actively engaged on the platform.

Outlier Analysis

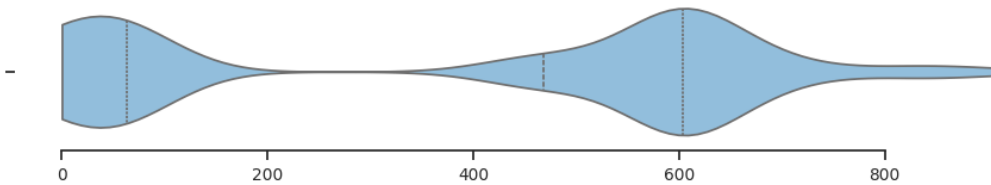
Distribution Density & Outlier Spread: age



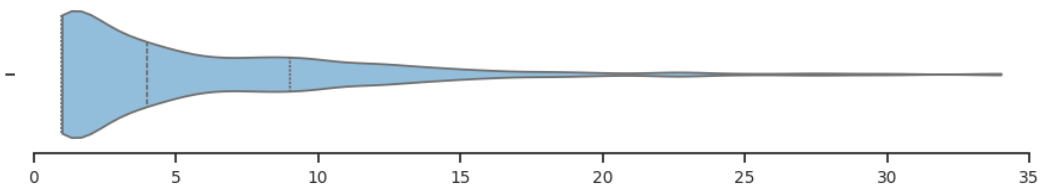
Distribution Density & Outlier Spread: days_to_apply



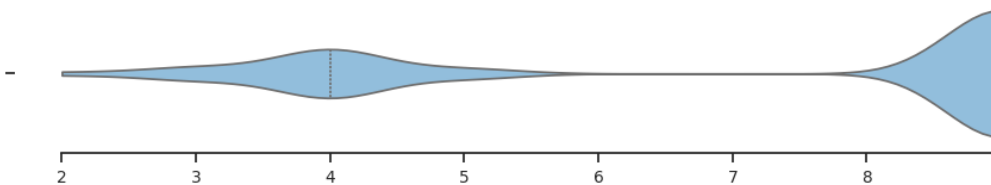
Distribution Density & Outlier Spread: engagement_duration_days



Distribution Density & Outlier Spread: participation_frequency



Distribution Density & Outlier Spread: total_category_opportunities



Key Insights and Explanation

Age: The data stays tightly grouped around 23 years old. A few older users reach up to 50 years, but these cases are rare and do not distort the overall picture, meaning no special cleanup is needed.

Days to Apply: Most people apply within 10 days, but a small group takes up to 420 days. This shows that some users keep their accounts inactive for over a year before finally taking action, proving that inactive users are not always lost leads.

Participation Frequency: A tiny segment of users interacts up to 34 times, compared to the normal range of 1 to 5 times. These power users show true high activity rather than data errors, so they should be grouped separately to avoid pulling up normal averages.

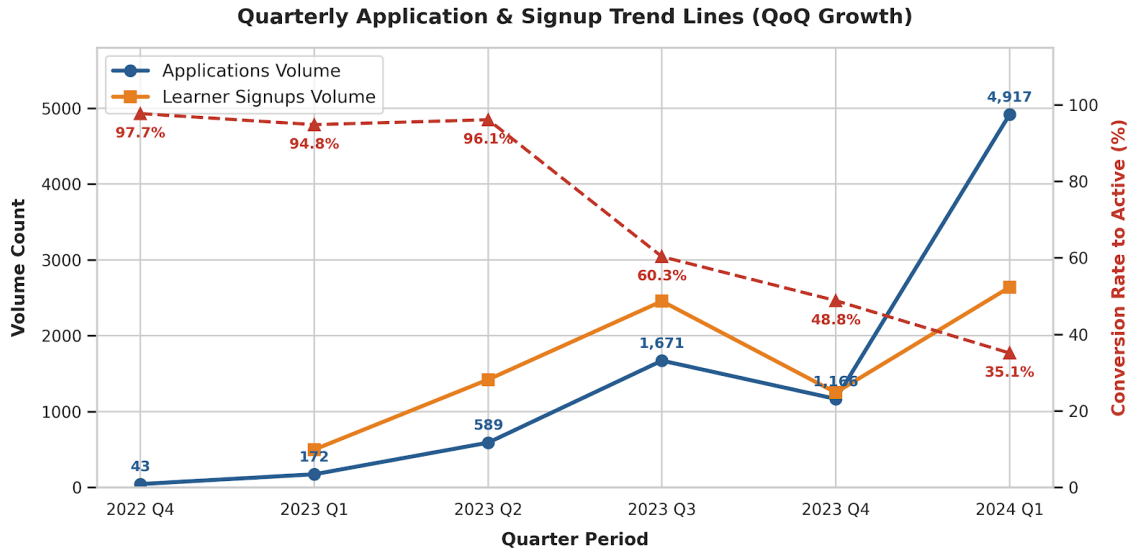
Engagement Duration Days: User lifespans split sharply into two distinct groups: those who leave before 100 days and those who stay for around 600 days. This reveals a critical early window where users either drop off quickly or become highly loyal long-term members.

Total Category Opportunities: User choice stops sharply at fixed limits of 4 and 9 options. This hard cutoff shows that system rules or plan restrictions are controlling user choices, rather than users choosing freely on their own.

4. Quarterly Trend Analysis

Applications per Quarter

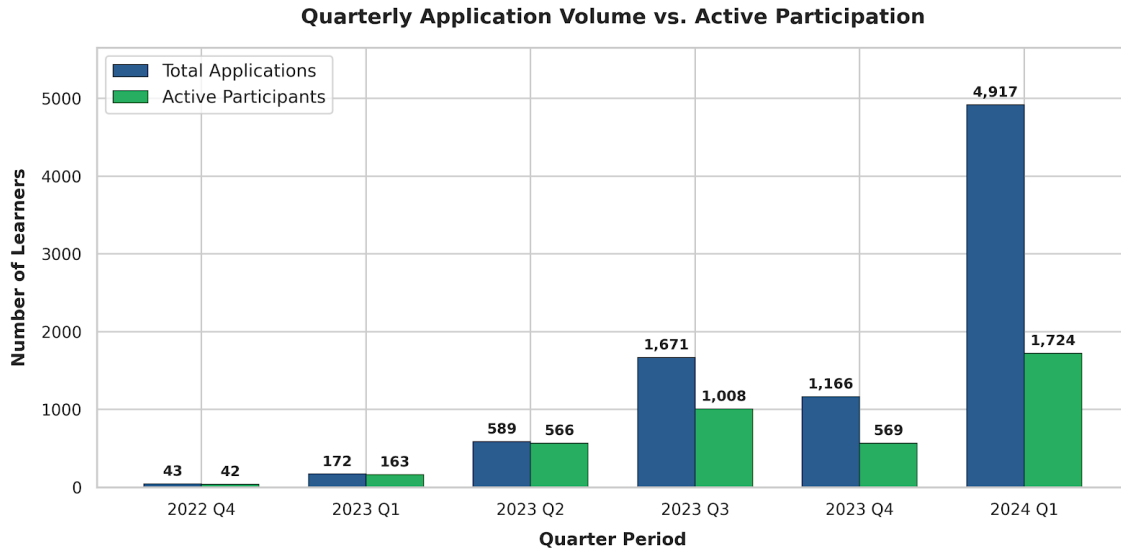
A total of 8,558 applications were recorded between 2022 Q4 and 2024 Q1. Application volume increased significantly throughout the analysis period, with 2024 Q1 recording the highest number of applications (4,917), contributing 57.5% of all applications. This was preceded by a temporary decline during 2023 Q4, where applications fell from 1,671 (19.5%) in 2023 Q3 to 1,166 (13.6%). Despite this decrease, application volume rebounded strongly in 2024 Q1, recording the highest quarter-over-quarter growth (321.7%) across the entire study period.



Looking at the quarterly application trend, 2024 Q1 clearly represents the peak application period, contributing more than half of all applications recorded in the dataset. To improve readability, the application trend was visualized using a grouped bar chart comparing Total Applications with Active Participants, allowing the growth in learner demand and participation to be compared directly across quarters.

Signups per Quarter

The dataset contains 8,558 learner signups between 2023 Q1 and 2024 Q1, with 295 records containing unspecified signup dates. Signup trends closely followed the application trend, with 2024 Q1 recording the highest number of signups (2,639), representing 30.8% of all learners. The largest quarter-over-quarter increase occurred during 2023 Q2, where signups increased by 185.3% compared to the previous quarter. Although signup volume declined by 49.1% during 2023 Q4, it recovered strongly with a 111.1% increase in 2024 Q1.



Overall, 2024 Q1 experienced the highest learner acquisition, while 2023 Q3 ranked second with 2,455 signups. The accompanying line chart was redesigned using clear quarterly labels and data labels to illustrate application growth, signup growth, and conversion rates within a single visualization.

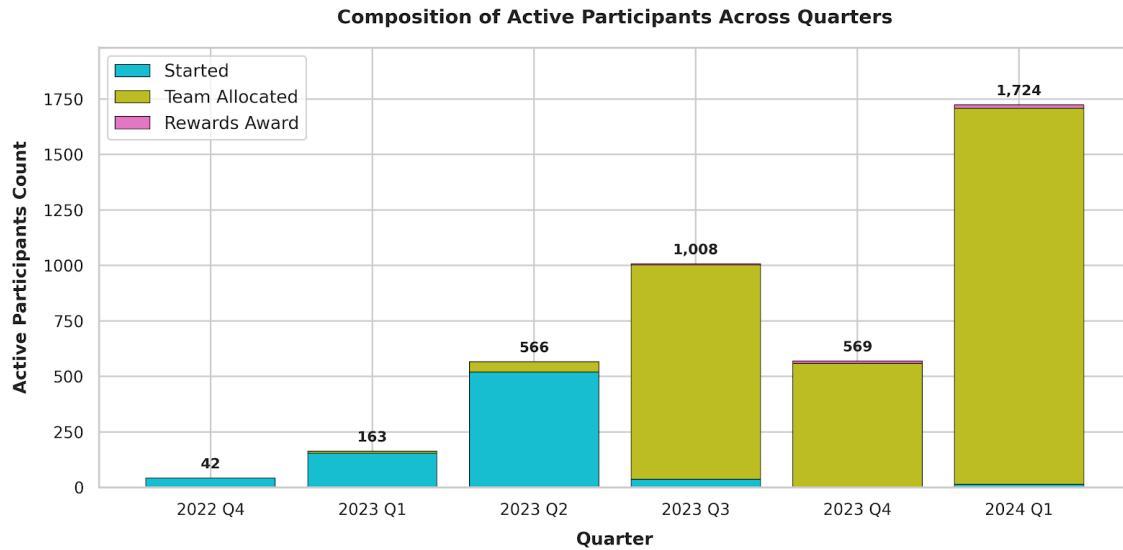
Participation per Quarter

Learner participation was evaluated using the three active participation statuses:

- Started
- Team Allocated
- Rewards Award

Together, these statuses represent learners who progressed beyond the application stage.

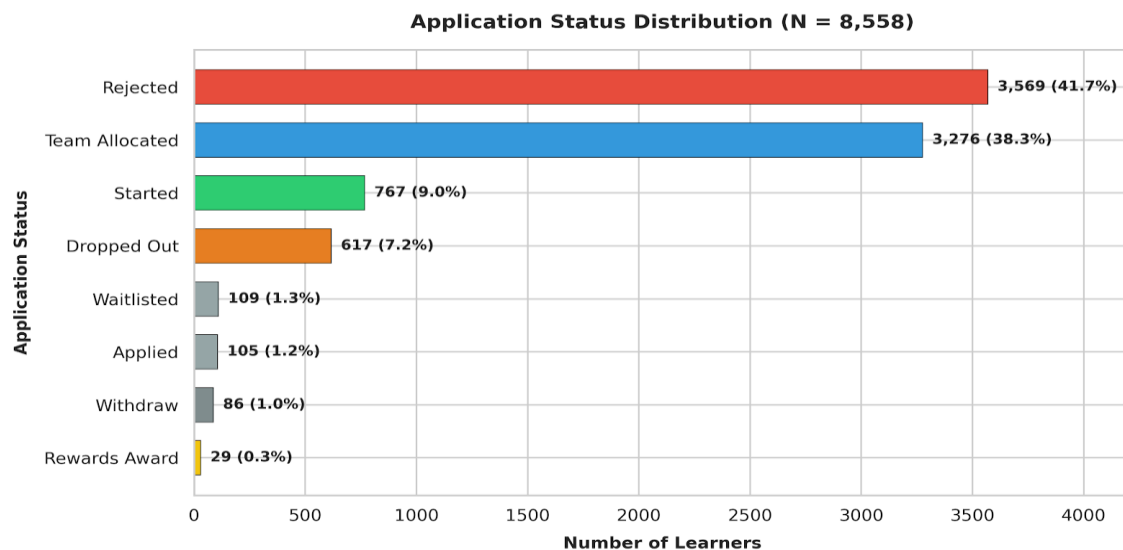
Of the 8,558 applications, 4,072 learners reached active participation. During the early quarters (2022 Q4–2023 Q2), conversion from applications to active participation remained consistently above 94%. However, beginning in 2023 Q3, conversion rates declined as application volumes increased substantially.



Although 2024 Q1 recorded the largest number of active participants (1,724 learners), it also experienced the lowest conversion rate (35.1%). This suggests that application growth outpaced available programme capacity, resulting in a smaller proportion of applicants progressing to active participation.

Application Status Distribution

To better understand learner progression, application outcomes were analysed across all status categories.



The analysis indicates that Team Allocated became the dominant successful outcome as application numbers increased, while Rejected became the largest individual status overall. Rejections increased dramatically during 2024 Q1, accounting for 2,770 learners, whereas Waitlisted learners only appeared during the same quarter. Rewards Award remained the least common successful outcome throughout the analysis period.

Monthly Trends

Monthly analysis was performed to identify seasonal patterns in learner behaviour.

Applications

Application activity increased steadily during 2023 before reaching its highest level in early 2024.

Peak Months	Applications
January 2024	2,431
February 2024	1,828
August 2023	760
December 2023	658
March 2024	658

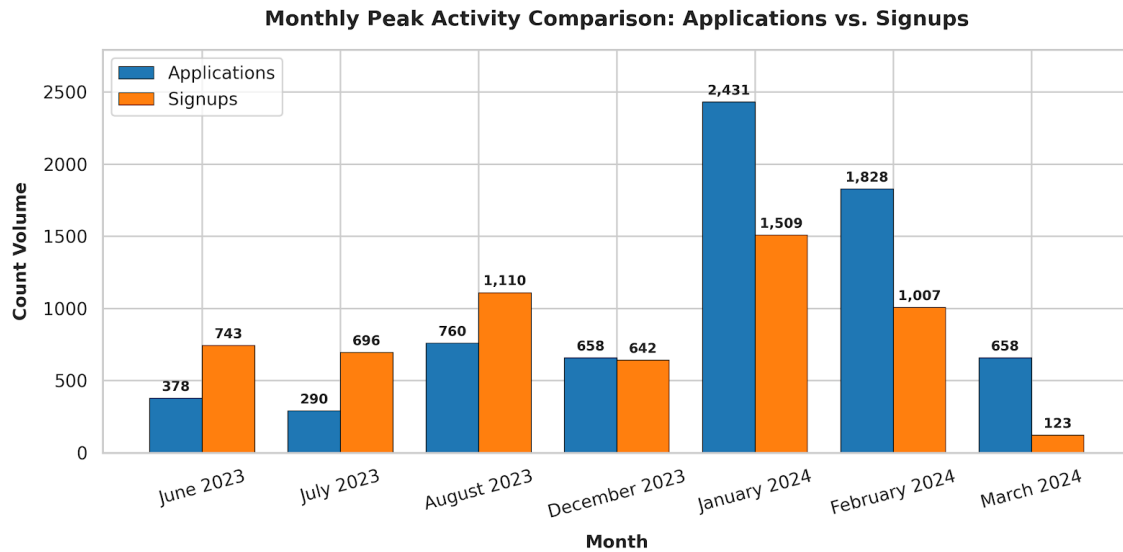
January 2024 alone contributed approximately **28%** of all applications, indicating a major recruitment or marketing cycle during the beginning of the year.

Signups

Signup activity followed a similar seasonal pattern.

Peak Months	Signups
January 2024	1,509

August 2023	1,110
February 2024	1,007
June 2023	743
July 2023	696



The strong alignment between signup and application peaks suggests that learner acquisition campaigns translated effectively into application activity.

Peak and Low Peak Seasons

Based on the quarterly analysis, the following periods were identified:

Peak Signup Period

- 2024 Q1
- 2,639 signups

Lowest Signup Period

- 2023 Q1
- 498 signups

Peak Application Period

- 2024 Q1
- 4,917 applications

Lowest Application Period

- 2022 Q4
- 43 applications

These findings indicate that learner interest and recruitment activity were strongest during the first quarter of the year, particularly in January and February.

Summary of Key Insights

- **Application Growth:** Total applications increased exponentially from 43 applications in 2022 Q4 to 4,917 applications in 2024 Q1, representing the highest quarterly application volume.
- **Signup Growth:** Learner signups followed a similar trend, increasing from 498 signups in 2023 Q1 to 2,639 signups in 2024 Q1, demonstrating strong learner acquisition.
- **Participation Growth:** Active participation increased from 42 learners in 2022 Q4 to 1,724 learners in 2024 Q1, although participation growth was slower than application growth.
- **Declining Conversion Rates:** Conversion from application to active participation declined from 97.7% during the earliest quarter to 35.1% in 2024 Q1, suggesting that applicant demand exceeded programme capacity.
- **Increasing Competition:** Rejections increased substantially in later quarters, with 2,770 rejected applications recorded during 2024 Q1, making rejection the largest application outcome.

- **Seasonality:** January and February represented the strongest recruitment period for both applications and signups, while 2023 Q4 experienced a temporary slowdown before a strong recovery in 2024 Q1.
- **Operational Shift:** Early quarters were dominated by learners with a Started status, whereas later quarters showed a clear transition towards Team Allocated, reflecting changes in programme operations and learner management.

Overall, the quarterly trend analysis demonstrates rapid platform growth in both learner signups and applications. However, as demand increased, conversion rates declined, and rejection rates rose, indicating that programme capacity did not expand at the same pace as learner interest. These findings suggest that future programme planning should focus on increasing capacity, improving conversion efficiency, and supporting higher learner participation while maintaining programme quality.

5. Opportunity Category Analysis

Methodology

Data source: processed_data.csv (8,558 rows, 34 columns). Analysis was performed in Python using pandas for data manipulation and matplotlib/seaborn for visualization. Apply dates were parsed to extract year, month, quarter, and weekday; participant identity was derived from first_name + institution_name. The complete, runnable script is included as a companion file (full_analysis.py).

The underlying dataset contains 8,558 application records. Apply dates were parsed to derive year, month, quarter, and weekday fields; 307 records (3.6%) had missing apply dates and were excluded from the time-based analyses (growth and seasonality) but retained in the category and distribution analyses.

Applications per Category

Objective

Measure total application volume across categories using both absolute counts and percentage share to identify the categories contributing most to overall platform activity.

Category	Applications	% of Total
Internship	5421	63.3%
Course	2037	23.8%
Event	545	6.4%
Competition	425	5.0%
Engagement	130	1.5%

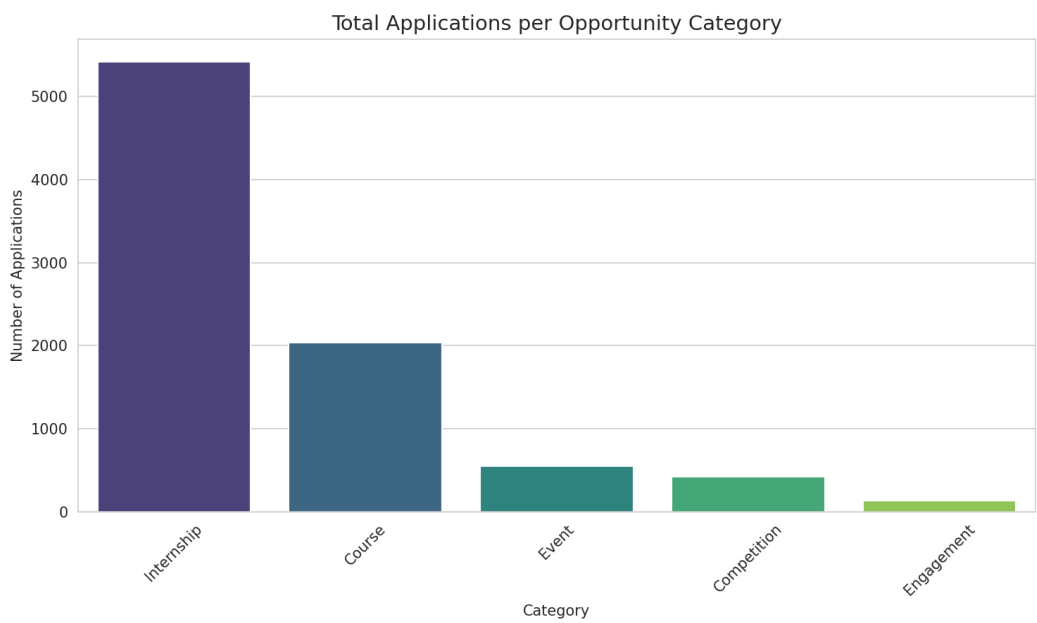


Figure 1. Total applications per opportunity category.

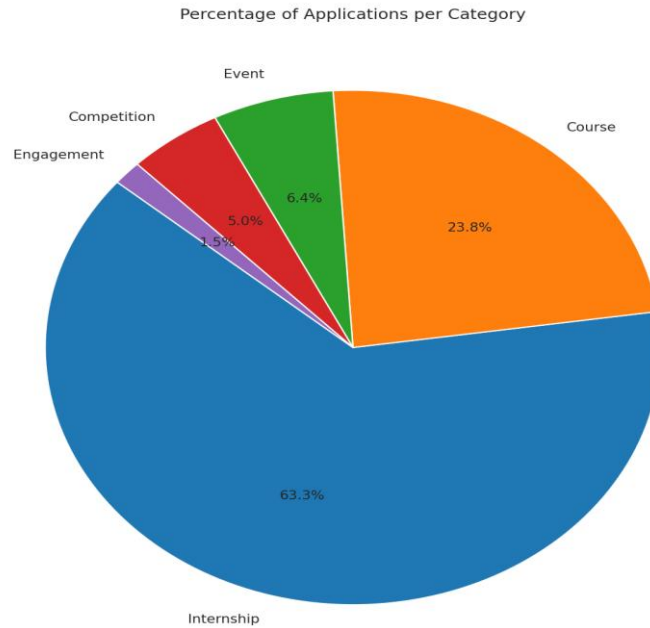


Figure 2. Share of applications by category.

Key Insights

Internship is the dominant opportunity category, accounting for 63.3% of all applications, followed by Course (23.8%). Together, these two categories represent 87.1% of total application volume, indicating that learner demand is highly concentrated in these opportunity types. In contrast, Event, Competition, and Engagement contribute a relatively small proportion of overall applications. These findings suggest that Internship and Course opportunities remain the primary drivers of platform activity.

6. Participation per Category (Unique Learners)

Measure the number of distinct learners participating in each category to evaluate category reach and determine whether application volumes are driven by broad participation or repeated submissions from the same learners.

Category	Unique Learners	Avg. Applications / Learner
Internship	2396	2.26

Course	1670	1.22
Event	441	1.24
Competition	254	1.67
Engagement	121	1.07

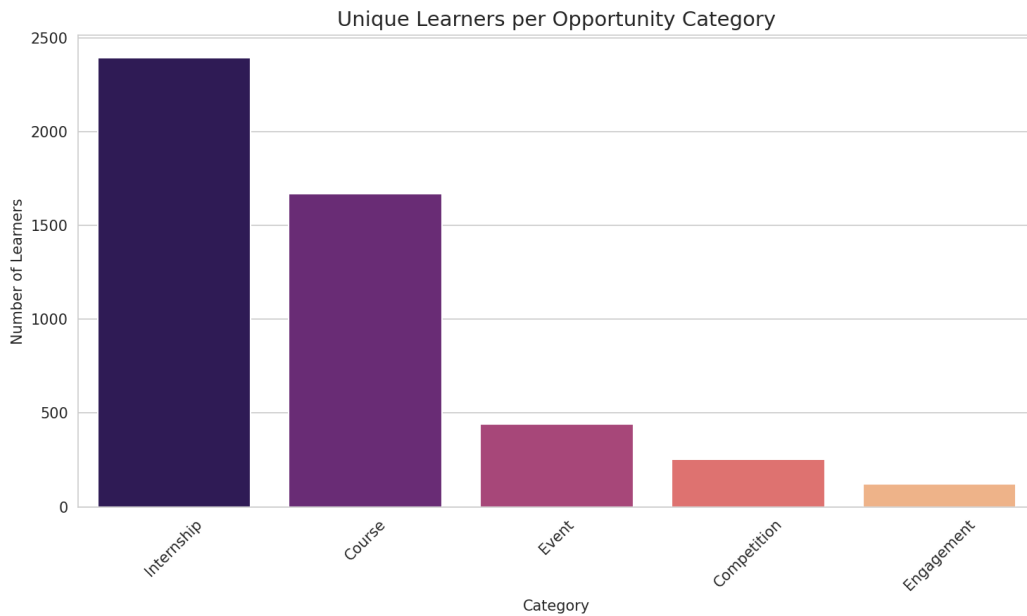


Figure 3. Unique learners engaging with each category.

Analysis of unique learner participation indicates that Internship attracts both the highest number of learners and the highest level of repeat engagement, with an average of 2.26 applications per learner. This suggests that the category's high application volume is driven by both broad participation and multiple applications from the same learners. In comparison, Course, Event, and Engagement average approximately one application per learner, indicating that participation in these categories is primarily characterised by single applications.

Category Growth

Analyze application trends by category over time, including year-over-year and quarterly comparisons, to identify growth patterns, declining interest, and seasonal fluctuations.

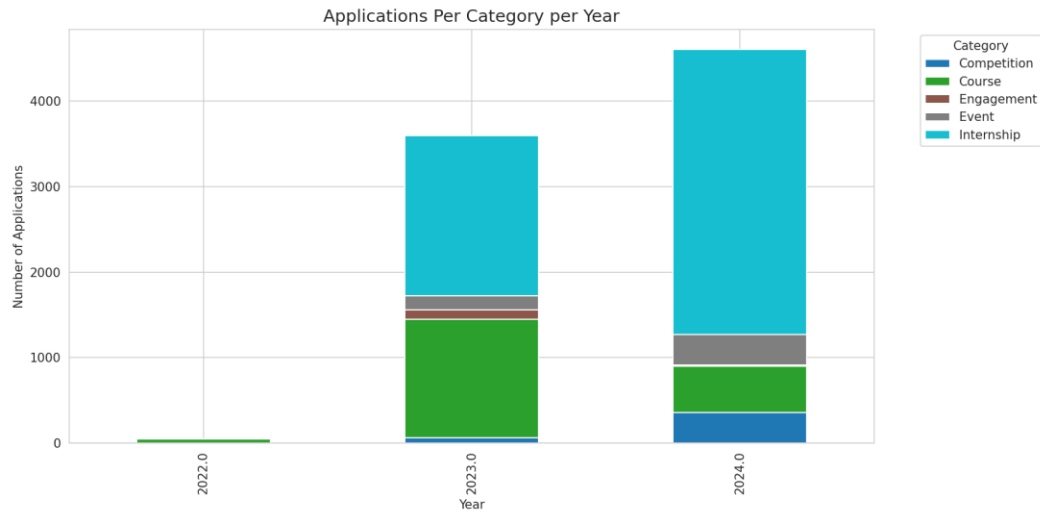


Figure 4. Applications per category, by year.

Category	2023	2024	YoY Change
Internship	1878	3340	+77.8%
Course	1391	540	-61.2%
Event	163	360	+120.9%
Competition	59	358	+506.8%
Engagement	107	12	-88.8%

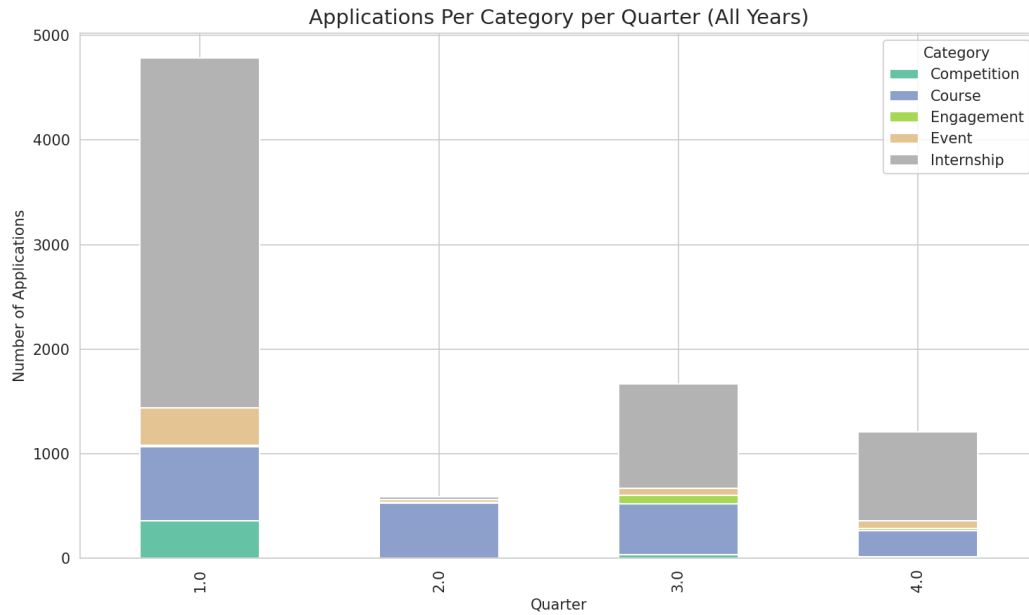


Figure 5. Applications per category by quarter, aggregated across all years.

Year-over-year analysis between 2023 and 2024 shows substantial growth in Internship (+77.8%), Event (+120.9%), and Competition (+506.8%) applications. Although Competition recorded the highest percentage increase, this growth should be interpreted with caution because it originated from a relatively small baseline. Conversely, Course (-61.2%) and Engagement (-88.8%) experienced significant declines, suggesting a reduction in learner demand or opportunity availability. Quarterly analysis further indicates that Q1 consistently records the highest application volumes, highlighting a recurring seasonal pattern that may inform future planning and resource allocation.

Seasonal Behavior

Evaluate application patterns across months and weekdays versus weekends to identify temporal trends, seasonality, and peak engagement periods within each category.

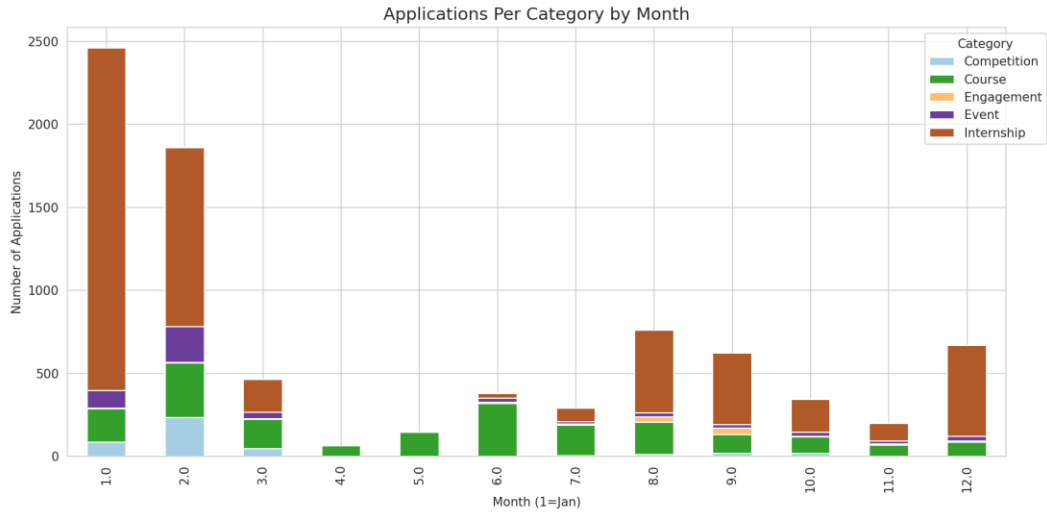


Figure 6. Applications per category by month (1 = January).

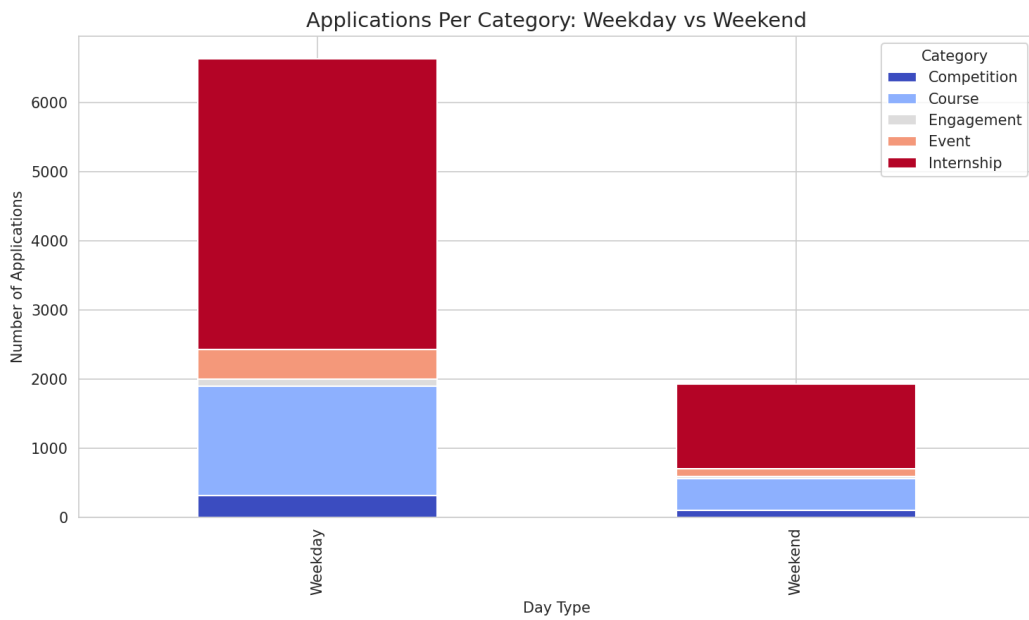


Figure 7. Applications per category, weekday vs. weekend.

Category	Weekday	Weekend	% Weekend
Internship	4203	1218	22.5%
Course	1579	458	22.5%
Event	426	119	21.8%
Competition	319	106	25.0%

Engagement	104	26	20.0%
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Application activity reaches its highest level in January, with 2,461 applications recorded during the analysis period. This pattern may reflect increased learner engagement at the beginning of the academic or calendar year. Additionally, 20–25% of applications are consistently submitted during weekends across all categories, indicating that application behaviour remains relatively stable throughout the week without a strong weekday or weekend preference.

Participation Distribution

Examine the distribution of applications per learner within each category to assess participation behavior, identify repeat applicants, and determine whether engagement is evenly distributed or concentrated among a smaller group of learners.

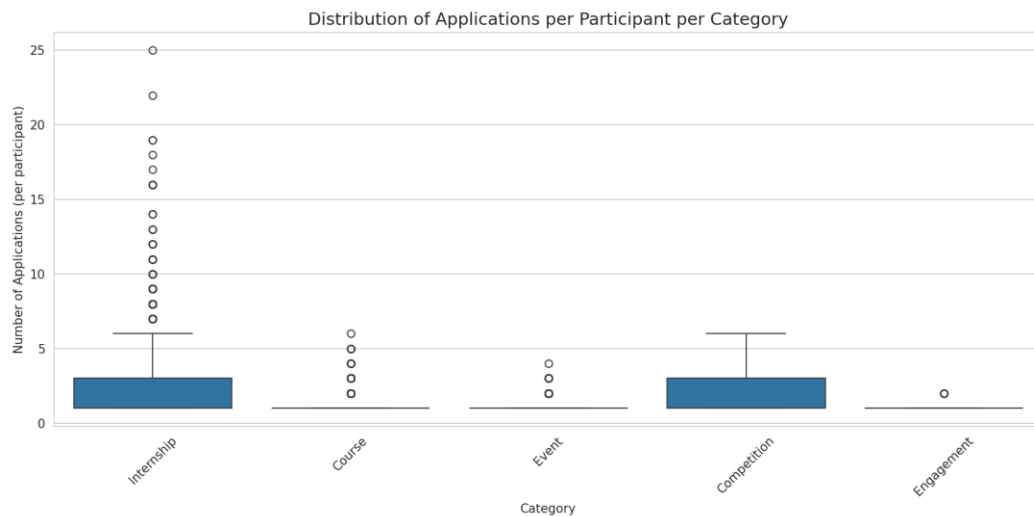


Figure 8. Distribution of applications per participant, by category.

Category	Median	Mean	Max	Std Dev
Internship	1	2.26	25	2.31
Competition	1	1.67	6	0.97
Event	1	1.24	4	0.51
Course	1	1.22	6	0.57
Engagement	1	1.07	2	0.26

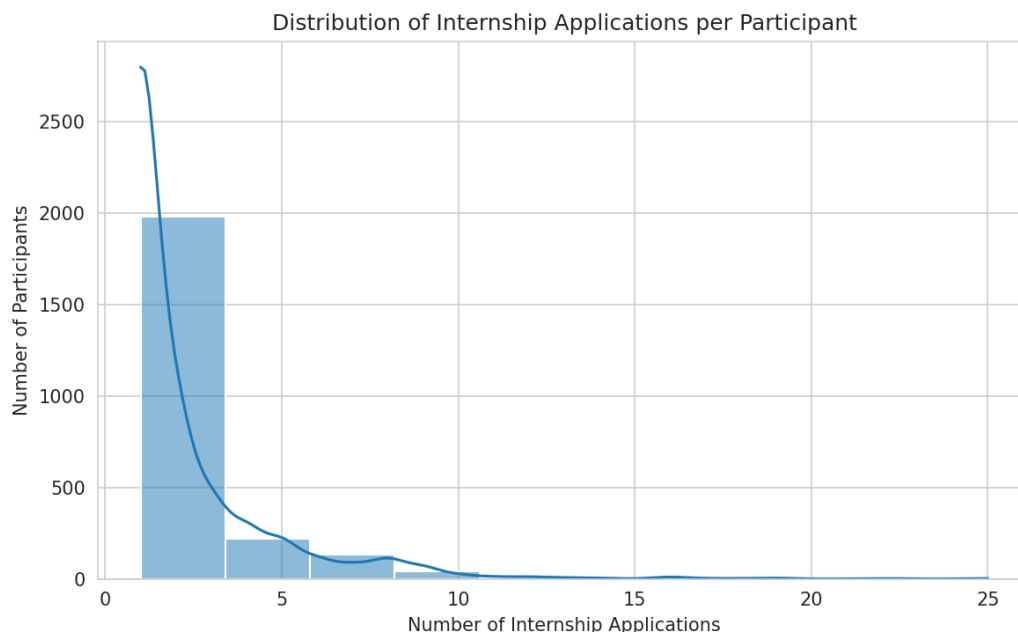


Figure 9. Distribution of Internship applications per participant (highest-volume category).

The median number of applications per learner is one across all opportunity categories, indicating that most learners submit a single application. However, the Internship category demonstrates a markedly different participation pattern, with an average of 2.26 applications per learner and a maximum of 25 applications submitted by a single learner. The gap between the mean and median suggests a positively skewed distribution driven by a relatively small group of highly active repeat applicants. Other categories show limited evidence of similar repeat application behaviour.

7. Trend Visualizations

Engagement Duration is calculated as Opportunity End Date minus Opportunity Start Date, in days. Signup-to-Apply Gap is calculated as Apply Date minus Learner SignUp DateTime, in days. A small number of rows have negative or missing values for these fields due to inconsistent source dates; these rows are excluded only from the specific chart they would distort, and are otherwise left in the underlying dataset. The top 2% of

extreme values was trimmed in several charts (duration- and gap-based) so that boxplots and histograms remain readable.

Participation Trends

Participation is measured as the volume of learner entries (records created against an opportunity) and how that participation breaks down by outcome status.

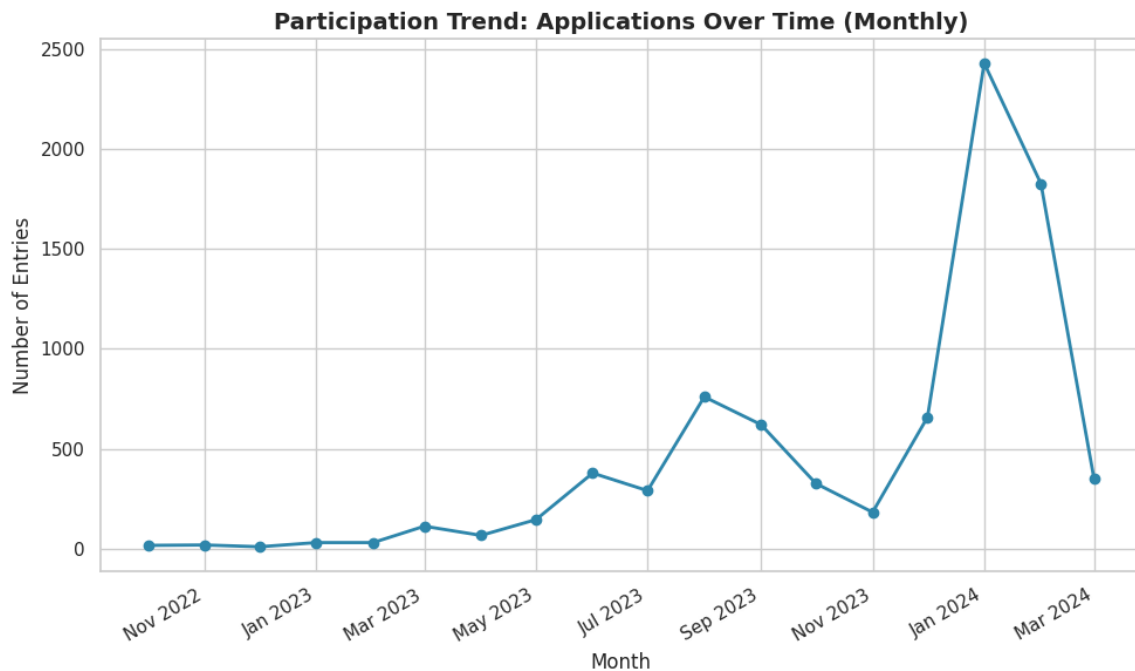


Figure 1. Participation Trend: Entries Created Over Time

What this shows: The monthly count of participation records created in the system. Peaks and dips reveal periods of high or low platform activity (e.g. cohort launches, seasonal drop-offs) and help identify whether overall participation is growing, flat, or declining over time.

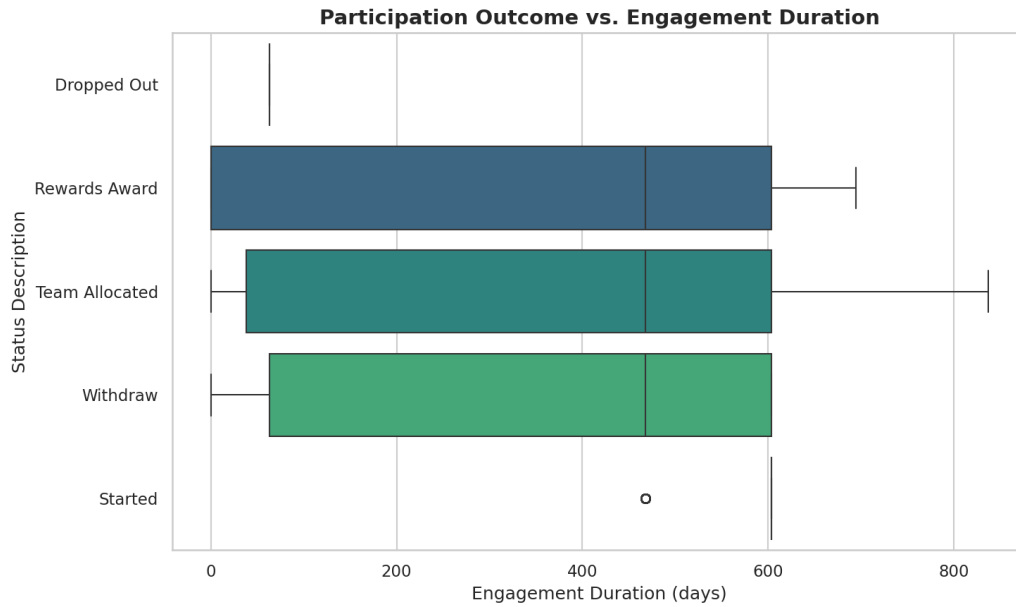


Figure 2. Participation Outcome vs. Engagement Duration (Boxplot)

What this shows: How long an opportunity ran in relation to the final participation status (e.g. Started, Rejected, Dropped Out, Team Allocated). This highlights whether longer-running opportunities are associated with different completion/drop-out patterns compared to shorter ones.

Signup Trends

Signup trends track when learners register — both the long-run trajectory and finer-grained timing patterns.



Figure 3. Signup Trend Over Time (Monthly) (Line Chart)

What this shows: The month-by-month volume of new learner signups. This is the core acquisition metric — rising trends suggest effective outreach periods, while sudden drops may flag broken funnels or reduced promotional activity.

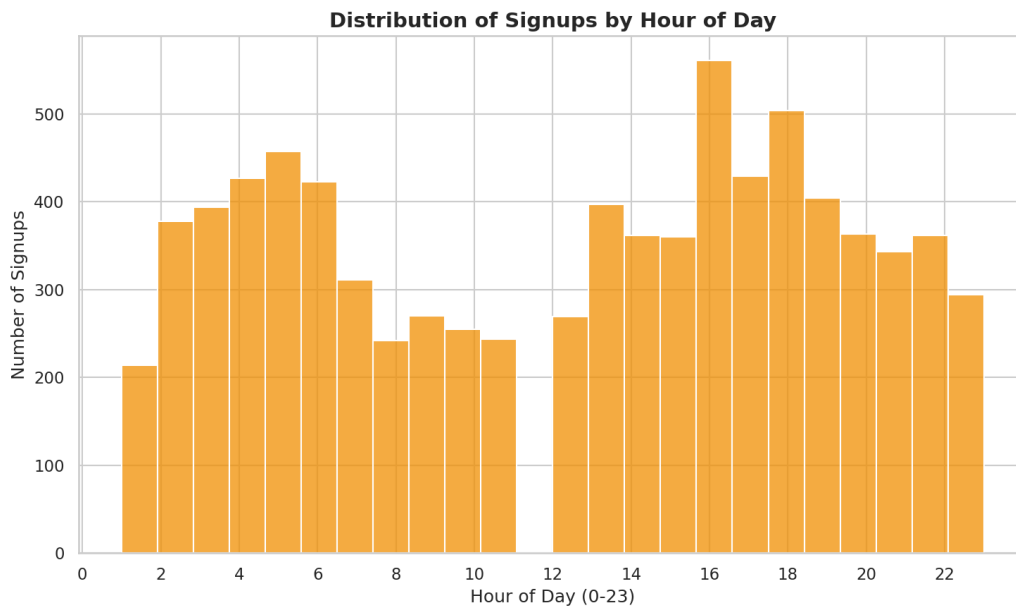


Figure 4. Distribution of Signups by Hour of Day (Histogram)

What this shows: What time of day learners tend to sign up. This is useful for scheduling communications, support availability, or platform maintenance windows around the hours of lowest activity.

Engagement Duration

Engagement duration is calculated as Opportunity End Date minus Start Date, in days — i.e. how long the opportunity a learner engaged with was scheduled to run.

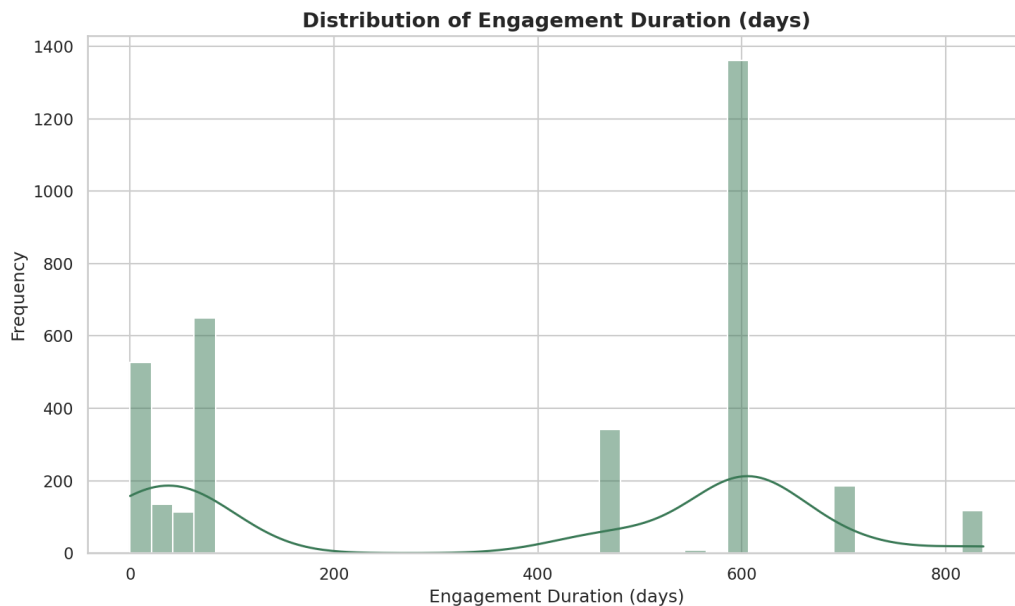


Figure 5. Distribution of Engagement Duration (days) (Histogram)

What this shows: The overall spread of opportunity lengths across the dataset. A right-skewed shape (common here) means most opportunities are short, with a smaller number of long-running programs.

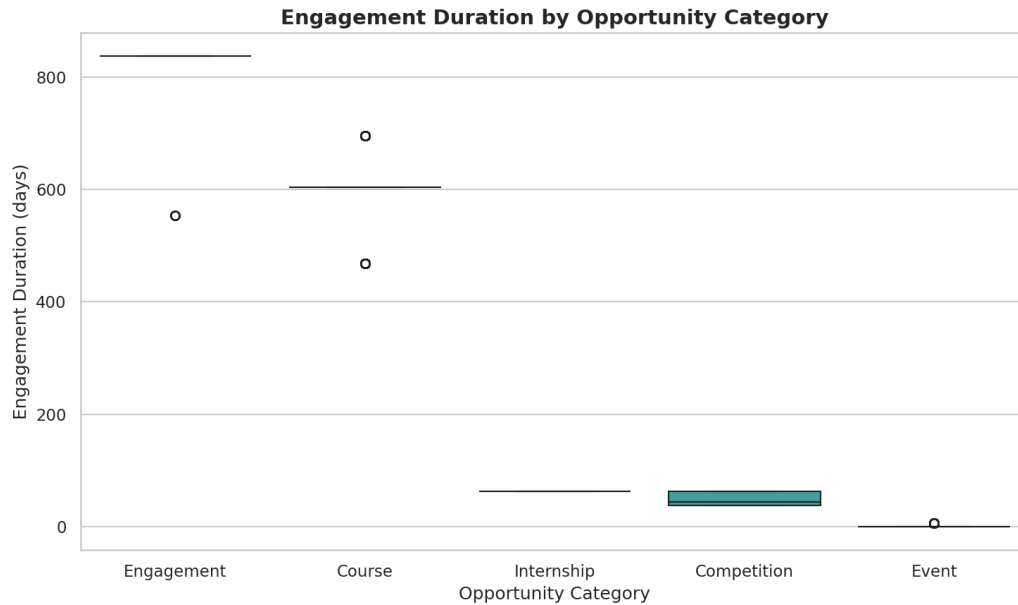


Figure 6. Engagement Duration by Opportunity Category (Boxplot)

What this shows: How engagement duration varies by opportunity type. For example, Internships typically run far longer than single Events or Courses — this chart quantifies that gap and flags categories with unusually wide variability.

Monthly Trends

A combined view of three key monthly funnel stages — signups, applications, and entries created — plotted together to compare their trajectories on one timeline.

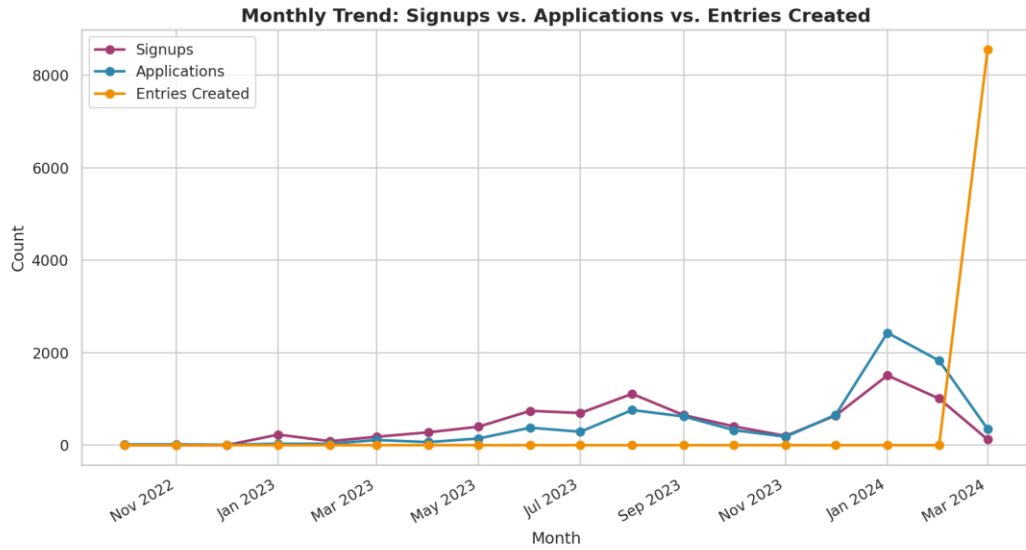


Figure 7. Monthly Trend: Signups vs. Applications vs. Entries Created (Line Chart)

What this shows: How the three funnel stages move together month-to-month. Gaps between the lines reveal drop-off (e.g. many signups but few applications), and parallel movement suggests the funnel is proportionally healthy across the period.

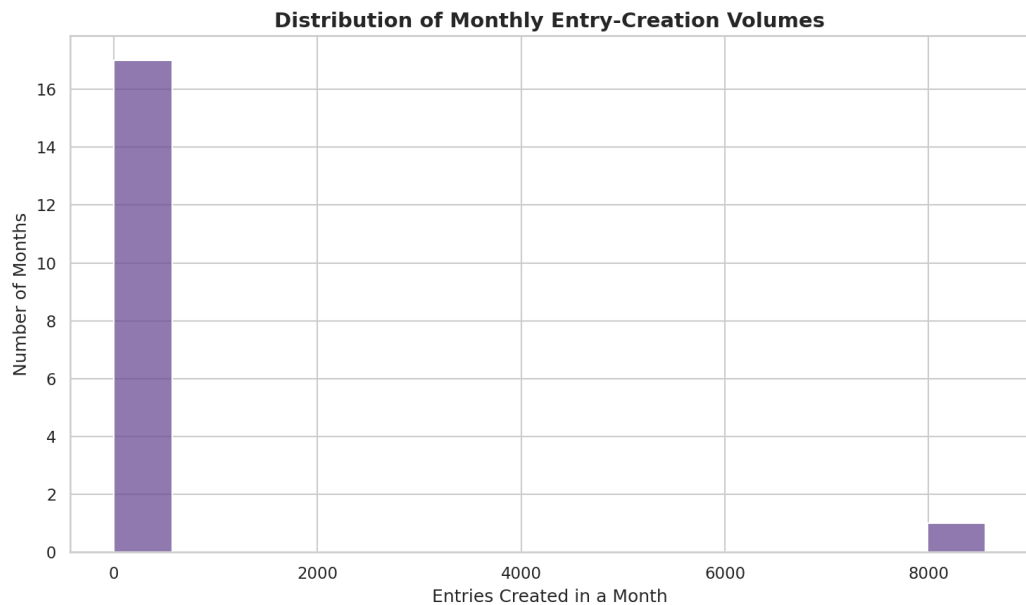


Figure 8. Distribution of Monthly Entry-Creation Volumes (Histogram)

What this shows: Rather than the time axis, this histogram looks at how many months fall into each volume range — helping identify whether most months are typical volume or whether a few outlier months are skewing the overall trend.

Opportunity Trends

Trends across the opportunities themselves — their category mix and how duration/timing relates to learner behaviour.

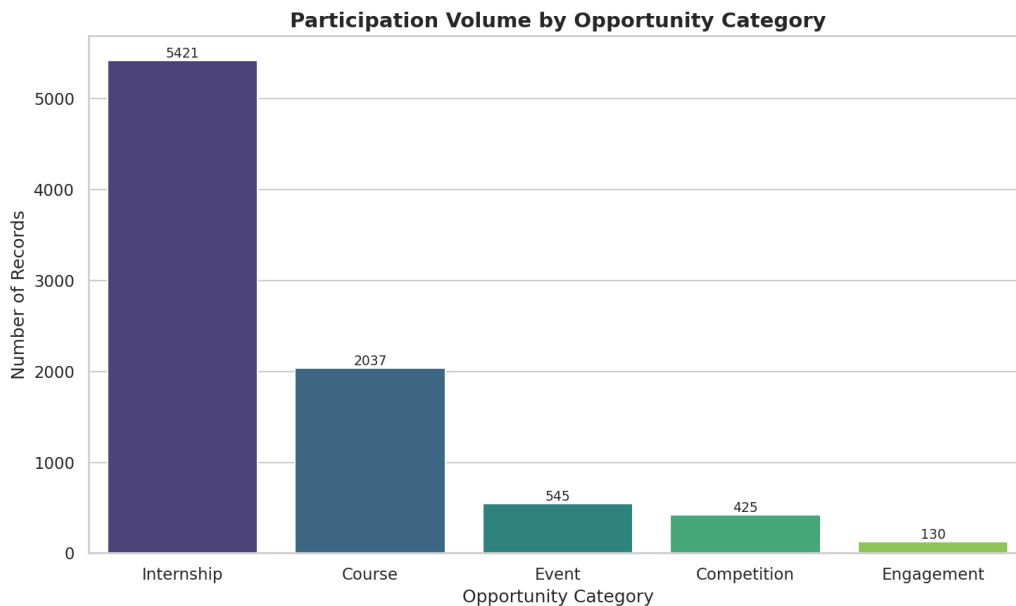


Figure 9. Participation Volume by Opportunity Category (Bar / Histogram)

What this shows: The relative popularity of each opportunity type. Internships dominate the dataset, followed by Courses, giving a quick read on where most learner activity is concentrated.

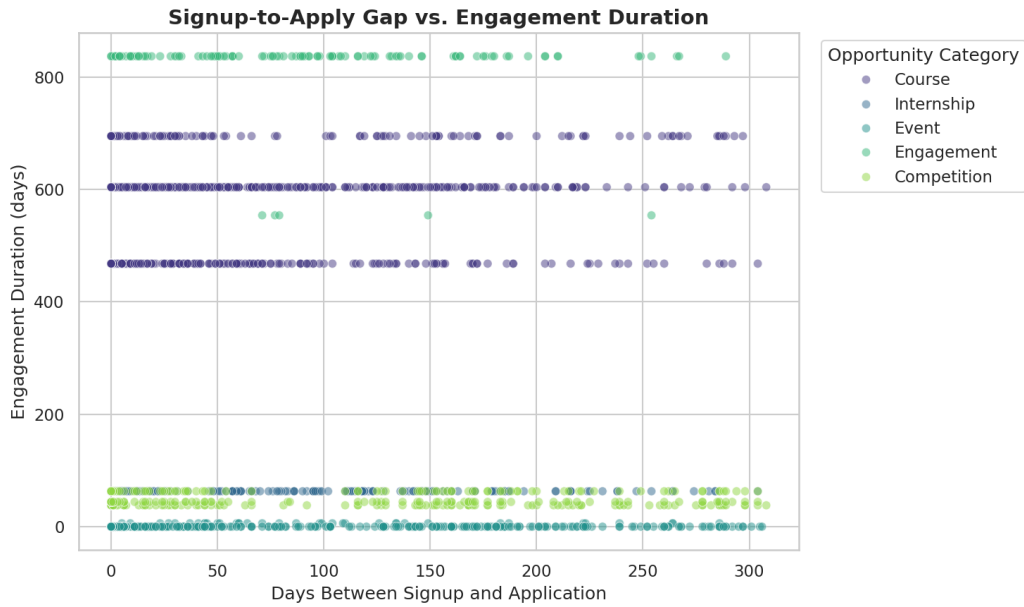


Figure 10. Signup-to-Apply Gap vs. Engagement Duration (Scatter Plot)

What this shows: Whether learners who take longer to apply after signing up tend to engage with longer- or shorter-running opportunities, and whether this pattern differs by category.

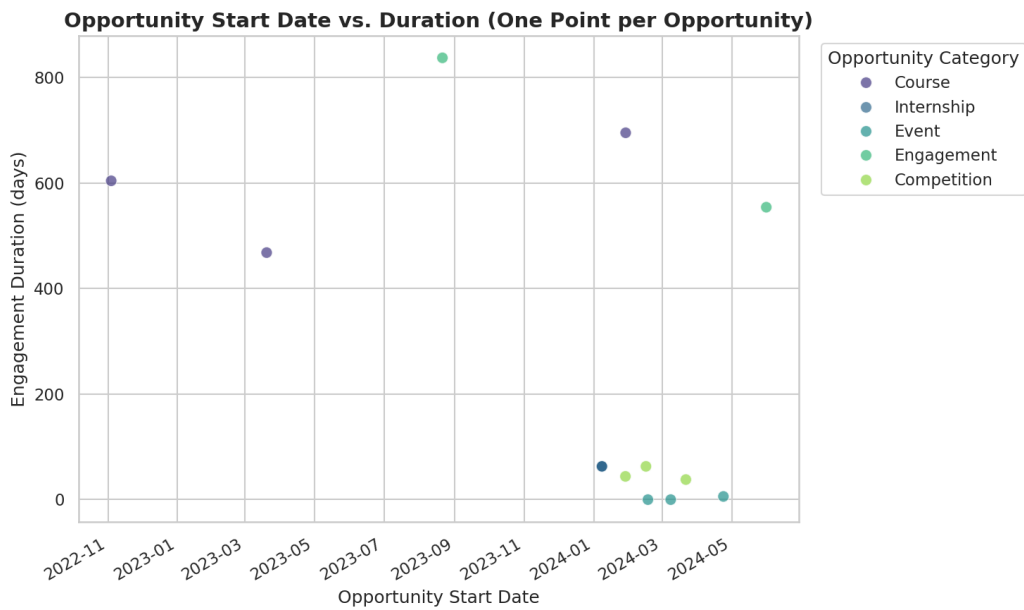


Figure 11. Opportunity Start Date vs. Duration (One Point per Opportunity) (Scatter Plot)

What this shows: A one-point-per-opportunity view of when opportunities launched and how long they ran. This distinguishes opportunity-design trends from the earlier participation-level charts, which counted every learner record.

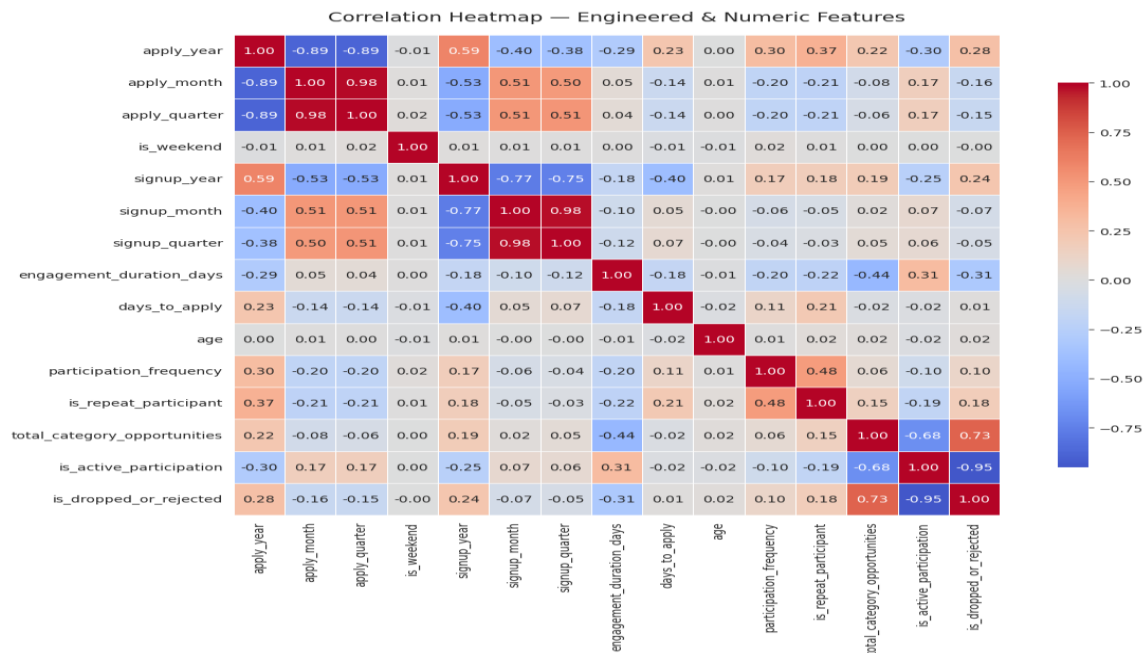
8. Heatmaps & Correlation Analysis

Methodology

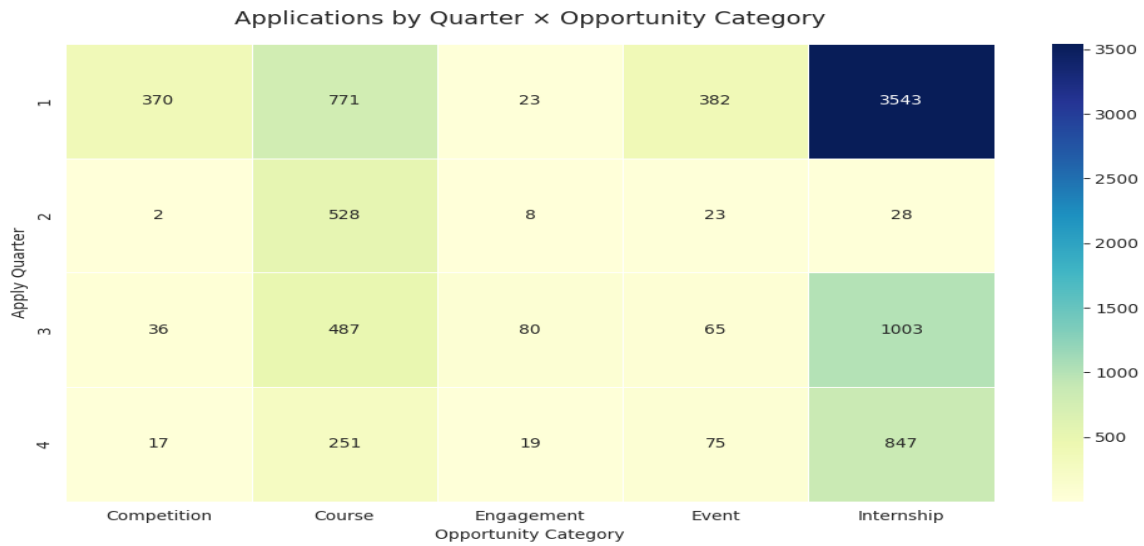
Using the `processed_data.csv` file (8,558 rows, 34 columns), I selected 15 numeric and engineered features — including date-derived fields (apply/signup year, month, quarter), engagement duration, days to apply, age, participation frequency, and outcome flags (active participation vs. dropped/rejected) — and computed a Pearson correlation matrix using pandas, visualized with a seaborn heatmap. I then built two pivot-table heatmaps: one showing raw application counts by Quarter $\times$ Opportunity Category (`pd.crosstab`), and one showing the average active participation rate for the same breakdown (`pivot_table` with mean aggregation).

Observations

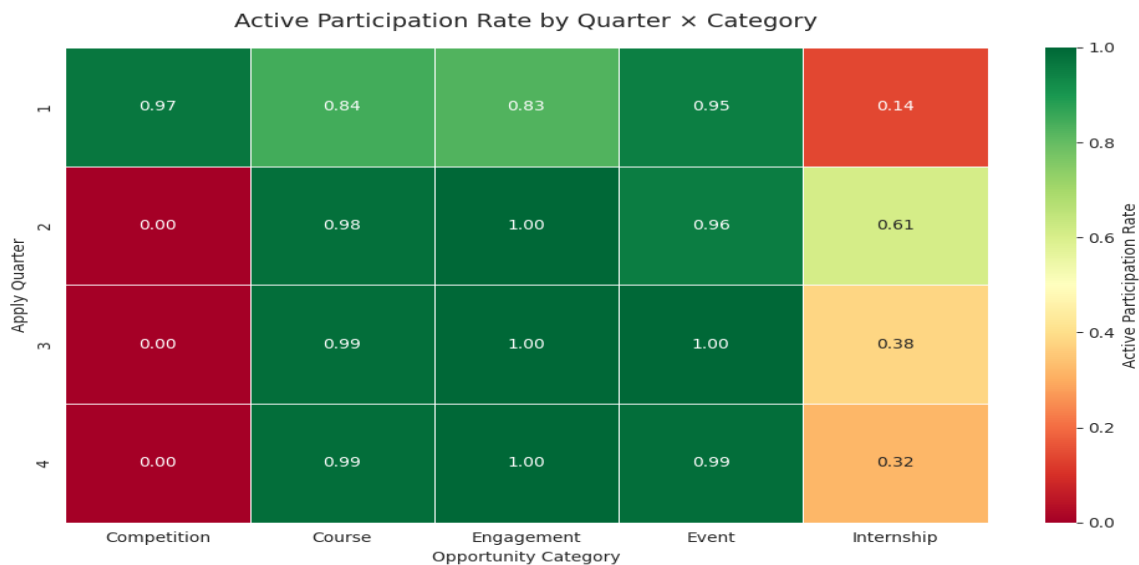
`is_active_participation` and `is_dropped_or_rejected` are strongly negatively correlated (-0.95), which is expected since they represent near-opposite outcomes.



total_category_opportunities showed the strongest relationship with participation outcomes among the engineered features (0.68–0.73), suggesting that categories with more available opportunities see different engagement behavior — a useful signal for prediction.



Quarter 1 accounts for the highest application volume overall, driven heavily by Internships (3,543 applications). Internships also show a notably lower participation rate in Q1 (0.14) compared to other categories, which sit at 0.83–1.00 across most quarters.



Date-part features (year/month/quarter) mostly correlate with each other since they're derived from the same timestamp — not independently useful for modeling.

Data Limitations

engagement_duration_days is missing for 59% of records (5,085 of 8,558), traced to missing Opportunity Start Dates for Rejected/Waitlisted applicants identified in Week 1. Correlations involving this feature reflect only the ~3,473 rows where it's available.

participation_frequency and is_repeat_participant are built using First Name + Institution Name as a proxy for identifying the same student, not a unique ID — this may slightly overcount repeat participation where students share a common name at the same institution.

Key Takeaways

The most modeling-relevant signal identified is total_category_opportunities, which shows a meaningfully stronger relationship with participation outcomes than other engineered features. The sharp contrast between Internship participation rates (as low as 0.14) and other categories (consistently 0.83+) suggests Opportunity Category itself is likely to be a strong predictor for Week 3's model. These heatmaps and the underlying correlation matrix are ready to inform feature selection going into predictive modeling.

9. Forecasting Signals

The main objective is to identify the strongest forecasting signals associated with learner participation, this includes looking at historical participation trends, comparing opportunity categories, evaluating engineered features, spotting early indicators of successful opportunities, and documenting everything in a structured way. The report also gives recommendations that can be applied directly when predictive modelling starts.

Forecasting Signals Methodology

The forecasting signal analysis was conducted using the team's processed dataset (processed_data.csv), which had already undergone data cleaning, standardization,

feature engineering, and exploratory data analysis. The analysis builds upon the completed analyses to identify variables that consistently influence learner participation and can support predictive modelling in Week 3. Descriptive statistics, grouped summaries, correlation analysis, and visualisations were used to identify meaningful forecasting signals, and each finding was interpreted from both statistical and business perspectives.

High-Performing Opportunity Categories

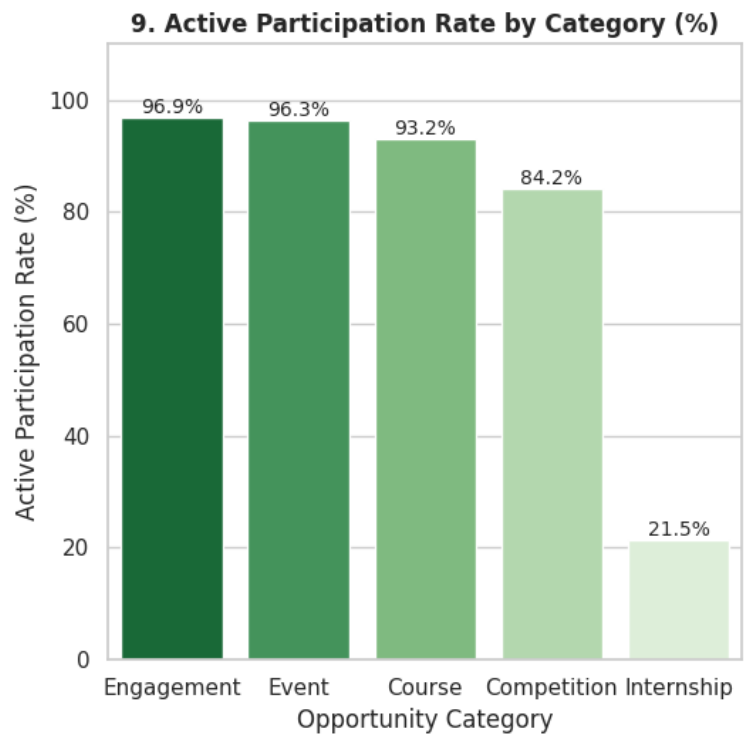


Figure 1: Active Participation Rate by Opportunity Category (%)

This chart was regenerated during the AI-assisted visualization review. The original version plotted several metrics (application volume, completion rate, active participation, and engagement) side by side on a single axis; because these metrics use different scales and units, it was difficult to read any one of them accurately. The chart below isolates a single, clearly-defined metric — active participation rate by category, sorted from highest to lowest with percentage labels on each bar — so the category ranking is unambiguous at a glance. Category performance is still one of the strongest predictors of future

participation, since learner interest tends to stay fairly consistent across similar opportunities.

Here, Engagement (96.9%), Event (96.3%), and Course (93.2%) have the highest active participation rates, while Internship sits far below at just 21.5%, despite being the largest category by far, with 5,421 entries, or 63% of all applications.

High-Performing Periods

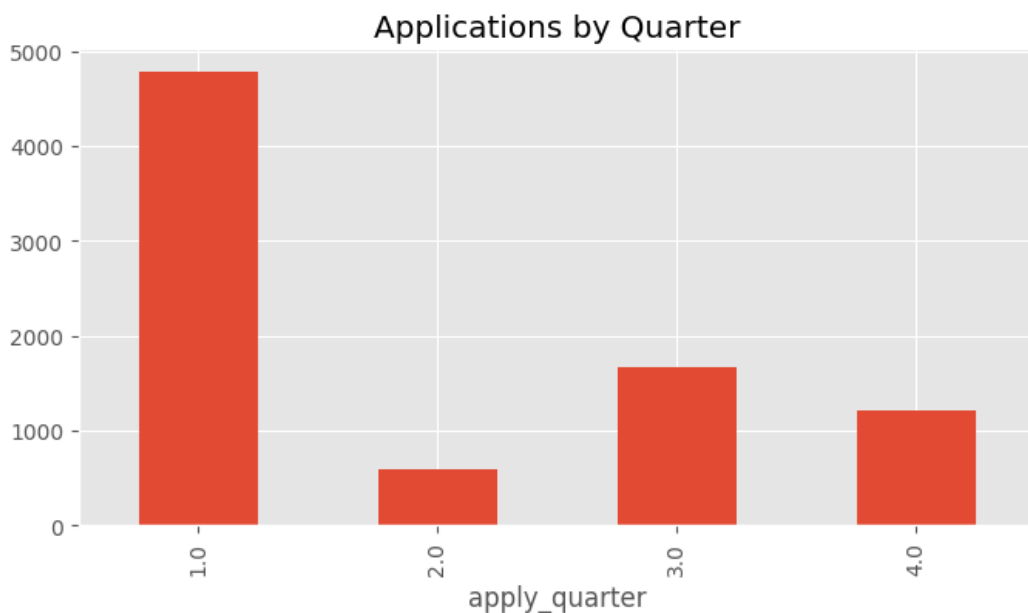


Figure 2: Applications by Quarter

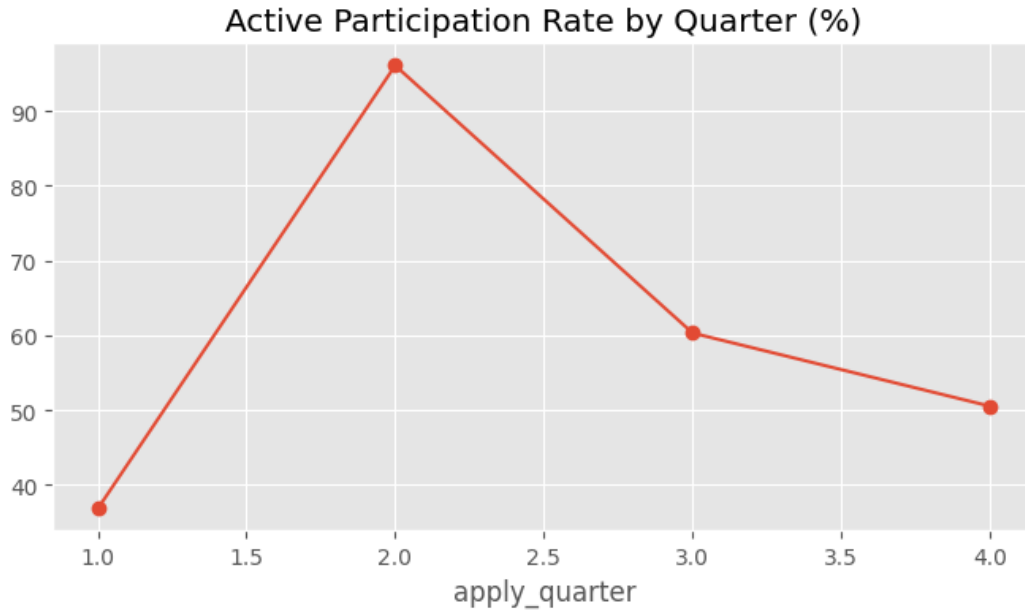


Figure 3: Active Participation Rate by Quarter (%)

Participation is analysed across months, quarters, and other available time intervals to identify seasonal peaks and periods of consistently high engagement. Once these periods are identified, possible reasons, like increased programme availability, academic schedules, or promotional activity, are worth discussing, since these periods become useful forecasting signals for similar future periods.

In our data, applications are heavily concentrated in Q1 (5,089 of 8,558 entries, about 59%), but Q1 actually has the lowest active participation rate at 37.1%. Q2, on the other hand, has the smallest volume (589 entries) but by far the highest active participation rate at 96.1%. Q3 sits at 60.3% and Q4 at 50.5%. So volume and success move in opposite directions here, the busiest quarter is also the toughest one to get accepted in.

Features Strongly Related to Participation

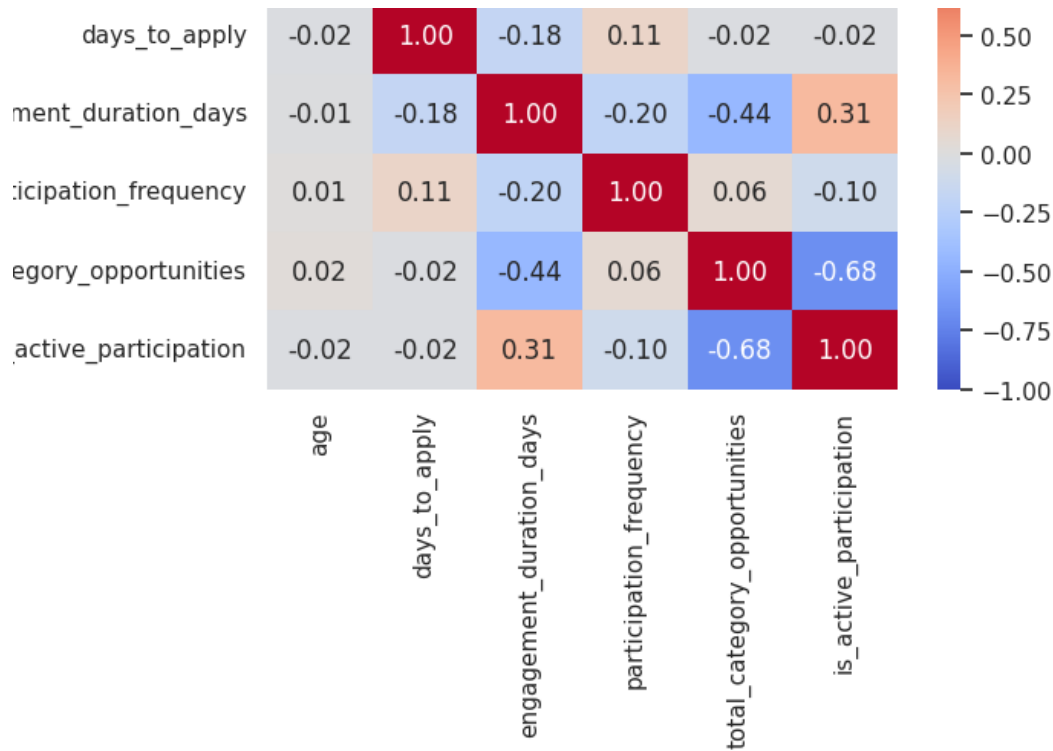


Figure 4: Correlation Heatmap of Core Numeric Features

This heatmap was regenerated during the AI-assisted visualization review. The original 15-variable version mixed date-part fields (apply_year, apply_month, signup_month, etc., which mostly just correlate with each other because they come from the same timestamp) alongside the outcome fields status_code and is_dropped_or_rejected, which restate the outcome itself rather than acting as independent drivers — that clutter made the genuinely useful relationships hard to spot. The version below keeps only the six core numeric/engineered features relevant to modelling: age, days_to_apply, engagement_duration_days, participation_frequency, total_category_opportunities, and is_active_participation.

In this focused view, total_category_opportunities has the strongest relationship with active participation ($r = -0.68$): learners exposed to fewer total opportunities within their categories are, on average, more likely to be actively participating.

Engagement_duration_days also shows a real positive relationship with active participation ($r = +0.31$) and a moderate negative relationship with

total_category_opportunities ($r = -0.44$). Participation_frequency and days_to_apply both show only weak relationships with the other features ($|r| \leq 0.20$), and age remains essentially uncorrelated with everything ($|r| \leq 0.02$), confirming it is not a useful predictive signal on its own.

Early Indicators of Successful Opportunities

Several early indicators can hint at whether a newly launched opportunity is likely to do well, strong registration activity in the first few days, higher repeat learner participation, shorter onboarding delay, a historically strong category, and lower dropout rates. Keeping an eye on these lets a team step in early and improve engagement before the programme wraps up.

Preparation for Predictive Modelling

The signals identified here form the analytical foundation for Week 3 predictive modelling. Features with a real relationship to participation should be selected as model inputs, and models like Logistic Regression, Decision Trees, Random Forest, or Gradient Boosting can then use them to predict future outcomes. Documenting this properly keeps feature selection transparent and easy to justify later.

Summary of Forecasting Signals

Forecasting Signal	Description	Business Importance	Expected Impact
High Performing Period	Time intervals with consistently higher participation.	Helps schedule future opportunities.	High
Opportunity Category	Categories attracting maximum learners.	Supports programme planning.	High
Participation Frequency	Measures learner engagement across opportunities.	Retention analysis.	High
Repeat Participant	Identifies returning	Measures loyalty.	High

	learners.		
Days to Apply	Evaluates onboarding speed.	Improves conversion.	Medium-High
Engagement Duration	Programme length.	Measures sustained engagement.	Medium
Active Participation	Successful learner participation.	Programme evaluation.	High

10. AI-Assisted Analysis

This analysis was generated using ChatGPT (OpenAI's AI assistant) applied to the team's processed dataset (processed_data.csv, 8,558 rows, 34 columns) along with the findings from the team's collective work during Week 2, which included data preprocessing, feature engineering, exploratory data analysis (EDA), quarterly trend analysis, opportunity category analysis, visualizations, and forecasting signals. Rather than repeating the descriptive analysis already completed, the AI was used to identify hidden patterns, explain observed trends, detect anomalies, suggest predictive features, and generate practical business recommendations. All AI-generated observations were reviewed against the processed dataset and the outputs of team members before being included in this report.

AI Prompts Used

Initial Prompt: “Analyze the provided dataset (processed_data.csv) and the accompanying summary of findings from the team’s exploratory data analysis. Focus on identifying non-obvious patterns, explaining the causes behind key trends, flagging any anomalies, and formulating practical business recommendations.”

Follow-up Prompt 1: “Based on the dataset, which are some non-obvious or hidden patterns that might not be immediately apparent from basic descriptive statistics? Please explore feature interactions and segment-specific behaviors”

Follow-up Prompt 2: “The team’s EDA identified that the ‘Internship’ opportunity category has high volume but a low success rate. Please provide a detailed explanation for this phenomenon and suggest potential strategies”

Follow-up Prompt 3: “Given the, which features do you recommend as the most important inputs for a predictive model in week 3? Justify your recommendations with evidence from the dataset”

Hidden Insights

Country differences are mostly explained by category mix, not nationality

The gap in participation success between countries (Ghana 61.8% vs. United States 40.0%) is largely a byproduct of what each country applies to, not a country-level effect on its own.

Country	% Applying to Internship	% Applying to Course
United States	70.4%	16.8%
India	61.8%	24.1%
Nigeria	57.5%	31.6%
Ghana	48.0%	36.7%

Countries with a higher share of Internship applications (the hardest category, 21.5% success rate) show lower overall participation success. Ghana's higher success rate is explained by applying to Course opportunities more often, not by being a stronger applicant pool.

Application speed has a non-linear relationship with participation success

apply_speed_category	Entries	is_active_participation Rate
1–3 Months	850	56.8%
Within 1 Month	806	51.9%
Same Day (<1d)	3,458	47.5%
3+ Months	2,032	45.6%

Not Recorded	534	43.1%
Within 1 Week	878	42.6%

Applicants in the 1–3 Months bucket have the highest participation success (56.8%), while Within 1 Week has the lowest (42.6%), lower even than Same Day applicants. This suggests a brief research/preparation period between signup and applying may improve applicant quality, whereas a rushed first-week application may reflect under-preparation.

Major choice matters more than category volume suggests

Information Systems is the most common major (2,173 applicants) but has a below-average `is_active_participation` rate (37.0%), while Computer Science, with half the volume (1,052), has a notably higher rate (50.0%). Applicant volume and applicant success are not correlated, the largest feeder major is not the most successful.

Repeat applicants succeed less often than first-time applicants

`is_repeat_participant` shows a clear negative relationship with outcome: first-time applicants succeed 64.0% of the time versus 42.0% for repeat applicants. This may indicate students who are unsuccessful on one application tend to apply again elsewhere, meaning repeat applications partly reflect prior rejection rather than stronger engagement.

Trend Explanations

The Internship category has the highest application volume (5,421 of 8,558 entries, approximately 63%) but also the lowest participation success rate (21.5%). This suggests that internships are highly competitive and oversubscribed, similar to real-world internship recruitment where available positions are limited compared to applicant demand.

`duration_category` shows a large gap: Short (<1 mo) and Multi-Year opportunities succeed at 99.6% and 98.9% respectively, while Unknown/Ongoing (5,085 entries, dominated by Internship) succeeds at only 18.5%, indicating opportunity duration and category are closely linked and duration alone is a strong signal.

Weekday signups outweigh weekend signups roughly 3.4-to-1 (6,631 vs 1,927), consistent with the userbase being students engaging with the platform during school/work weekdays rather than personal weekend time.

Anomaly Detection

Anomaly	Detail	Recommended Action
43 applications predate recorded signup	apply_date earlier than learner_signup_datetime for 43 rows	Flag as a timestamp logging inconsistency, not a behavioral anomaly
Internship funnel has no active late-stage status	All 5,421 Internship entries fall only into Rejected, Team Allocated, Dropped Out, Withdraw, or Applied, none reach a completion-stage status seen in other categories	Confirm whether Internship completions are tracked in a separate system
Very high-frequency applicants	144 entries belong to students with participation_frequency of 25 or higher, up to a maximum of 34 applications by one student	Review for bot-like or unusual account behavior vs. genuinely high engagement
Information Systems majors underperform their volume	Largest major by count, but below-average is_active_participation rate	Investigate whether this major is disproportionately tied to Internship applications

Business Recommendations

- Since Internship is the highest-volume yet lowest-success category, consider introducing an application readiness checklist or a short preparatory module before allowing internship applications. The analysis suggests that applicants who

spend approximately 1–3 months preparing before applying tend to achieve better participation outcomes.

- Review the 144 very high-frequency applicant entries (25+ applications, up to 34) to distinguish genuinely engaged students from potential bot or spam-like activity.
- For Information Systems and other high-volume, lower-success majors, consider steering students toward Course or Event opportunities where success rates are much higher (93–96%), to improve overall student outcomes without adding capacity.
- Clarify Internship's status tracking, since it never reaches a later completion stage, the platform may be missing visibility into actual internship completion and outcomes.

Suggested Predictive Features for Week 3 Modeling

Feature	Why It's Useful
opportunity_category	Strongest single predictor of participation outcome found across all analysis (21.5%–96.9% range)
apply_speed_category	Non-linear relationship with outcome; outperforms a raw continuous days-to-apply value
participation_frequency / is_repeat_participant	Captures both engagement and the repeat-applicant risk pattern (repeat applicants succeed less often)
country × opportunity_category interaction	Country alone is confounded by category mix; an interaction term captures the real effect
current_intended_major	Meaningful outcome variation independent of application volume
duration_category	Strong separation in outcome rate across duration buckets

AI Validations

The AI-generated insights presented in this report were validated against the processed dataset and the analytical outputs produced by the team during Week 2. AI was used as a decision-support tool to interpret patterns and generate recommendations rather than to replace statistical analysis or manual verification. The following validation steps were taken:

- Cross-referencing: All AI-generated patterns were cross-referenced with the team's EDA visualizations and statistical summaries.
- Consistency Checks: AI observations were checked for consistency with known domain knowledge and logical reasoning.
- Spot Verification: Key claims (e.g., success rates, volume counts) were manually verified against the raw data.
- Team Review: Insights were shared with team members for feedback and refinement before final inclusion

Key Insights

Overall, the AI-assisted analysis complemented the team's exploratory analysis by identifying hidden behavioural patterns, explaining relationships between key variables, detecting anomalies, and recommending practical improvements for the platform. It also highlighted important predictive features that can support the development of accurate machine learning models in Week 3. These insights provide a stronger understanding of learner participation behaviour and establish a solid foundation for predictive modelling.

Limitations

- The AI analysis is based on the processed dataset and the outputs generated during Week 2.
- Findings depend on the quality of the preprocessing and feature engineering completed by previous team members.
- The identified relationships are observational and should be further validated during Week 3 predictive modelling.

11. Additional AI-Assisted Visualizations

During the AI-assisted visualization review, the team also produced several additional charts from processed_data.csv that were not part of the original Week 2 chart set but that either confirm findings already discussed in this report in a clearer visual form, or surface new supporting detail (geographic spread, top institutions, application-speed and status breakdowns). They are included here as supplementary evidence rather than replacing the sections above.

Geographical Breakdown (Top 5 Countries)

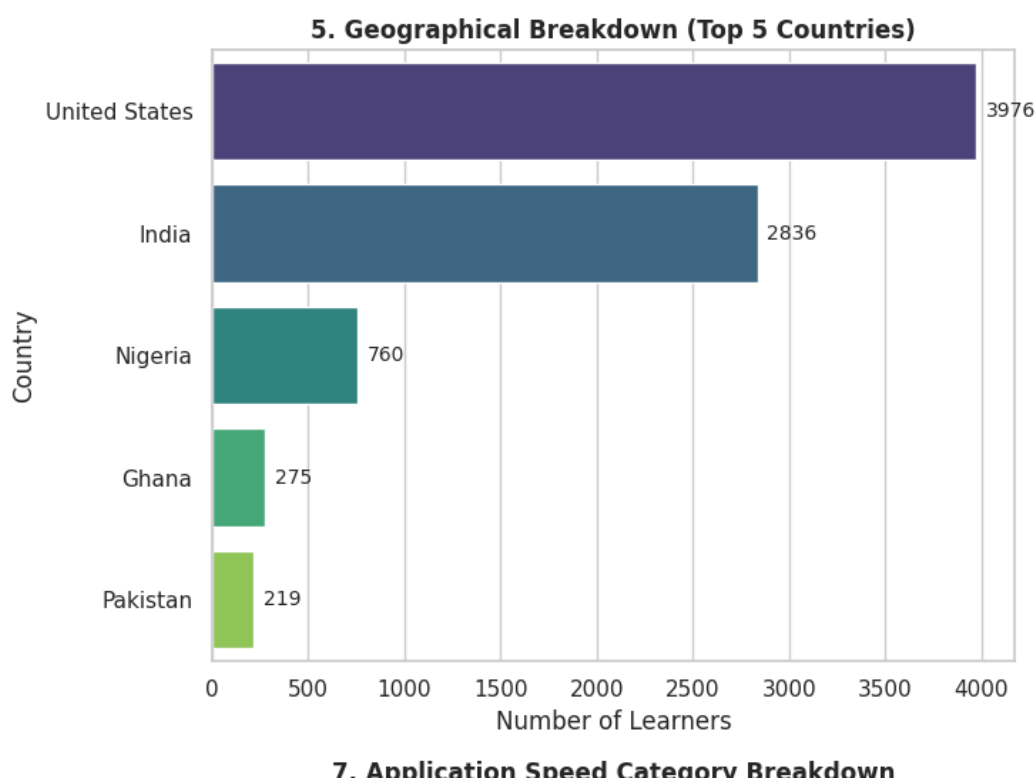
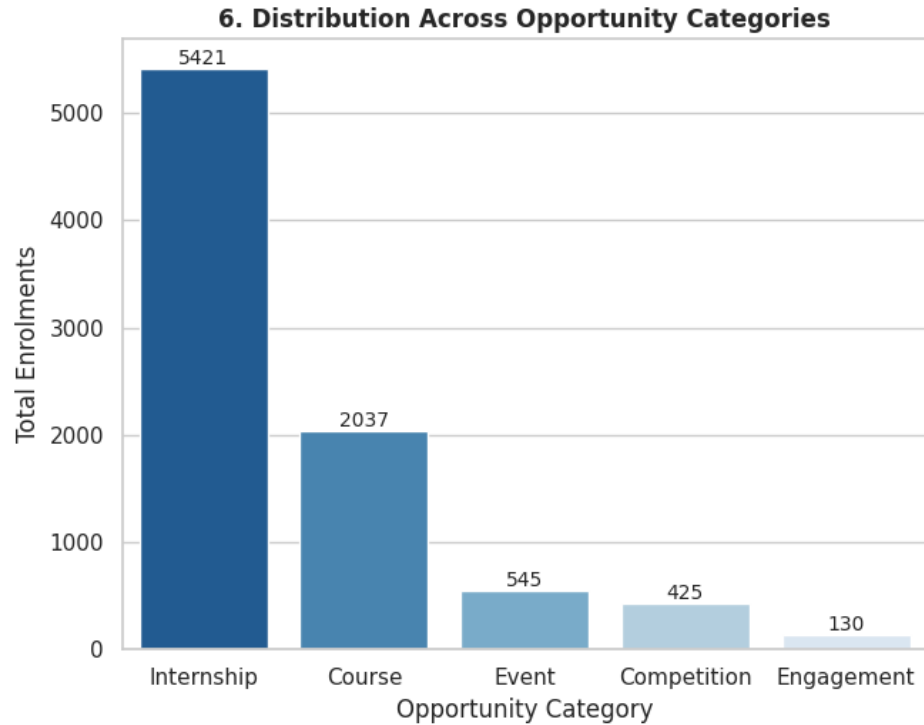


Figure: Geographical Breakdown (Top 5 Countries)

The United States (3,976 learners) and India (2,836) together account for the large majority of the platform's user base, followed by Nigeria (760), Ghana (275), and Pakistan (219). This matters for the country-level participation-rate differences discussed under Hidden Insights: country success rates differ mainly because countries apply to different mixes of categories, not because of the countries themselves, so any country-based modelling should account for this concentration.

Distribution Across Opportunity Categories



8. Top Learner Status Descriptions

Figure: Distribution Across Opportunity Categories

A cleaner, single-metric view of the category mix already reported in the Applications per Category table: Internship (5,421), Course (2,037), Event (545), Competition (425), and Engagement (130). Internship and Course together make up 87% of all applications.

Application Speed Category Breakdown

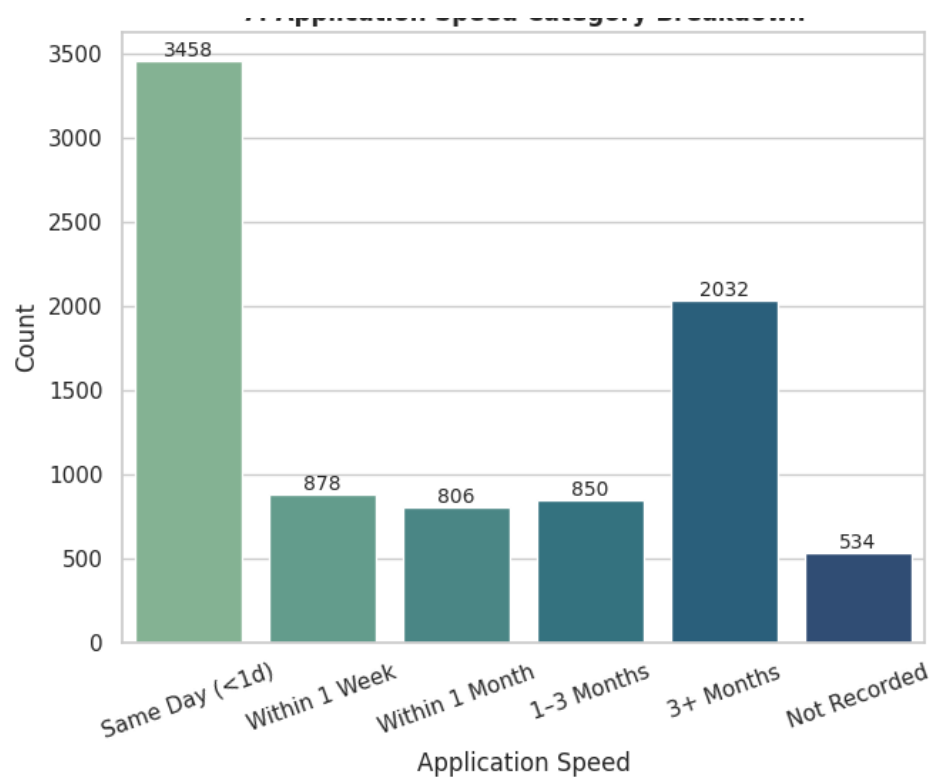


Figure: Application Speed Category Breakdown

Most applicants either apply the same day they sign up (3,458 entries) or wait 3+ months (2,032 entries) — a two-peaked pattern rather than a smooth spread. The remaining speed buckets (within a week, within a month, 1–3 months) are comparatively even, each in the 800–880 range.

Top Learner Status Descriptions

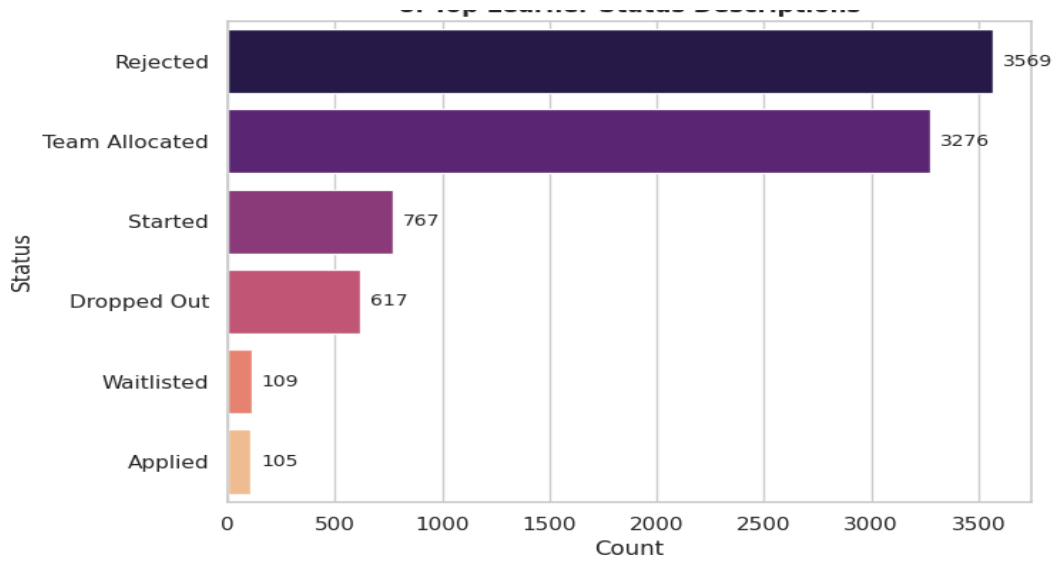


Figure: Top Learner Status Descriptions

Rejected (3,569) and Team Allocated (3,276) are by far the most common final statuses, together accounting for the large majority of records. Started (767) and Dropped Out (617) are next, while Waitlisted (109) and Applied (105) make up the smallest, still-in-progress segments.

Repeat vs First-Time Participant Volume

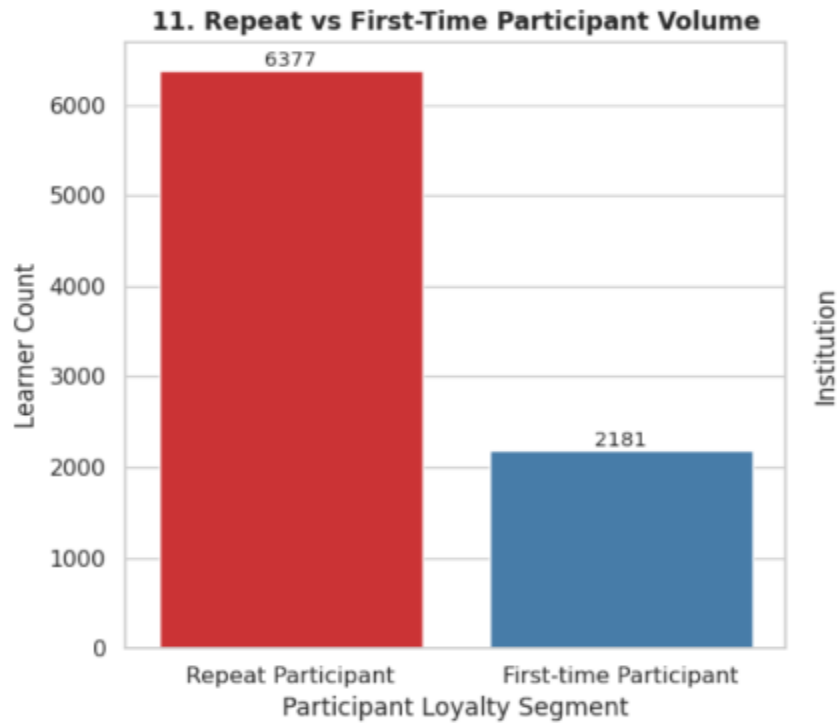


Figure: Repeat vs First-Time Participant Volume

Repeat participants (6,377) outnumber first-time participants (2,181) roughly 3-to-1 in this dataset. Combined with the finding under Hidden Insights that first-time applicants succeed more often (64.0%) than repeat applicants (42.0%), this suggests a sizeable share of repeat activity reflects re-applying after an earlier rejection rather than pure loyalty.

Top 5 Participating Educational Institutions

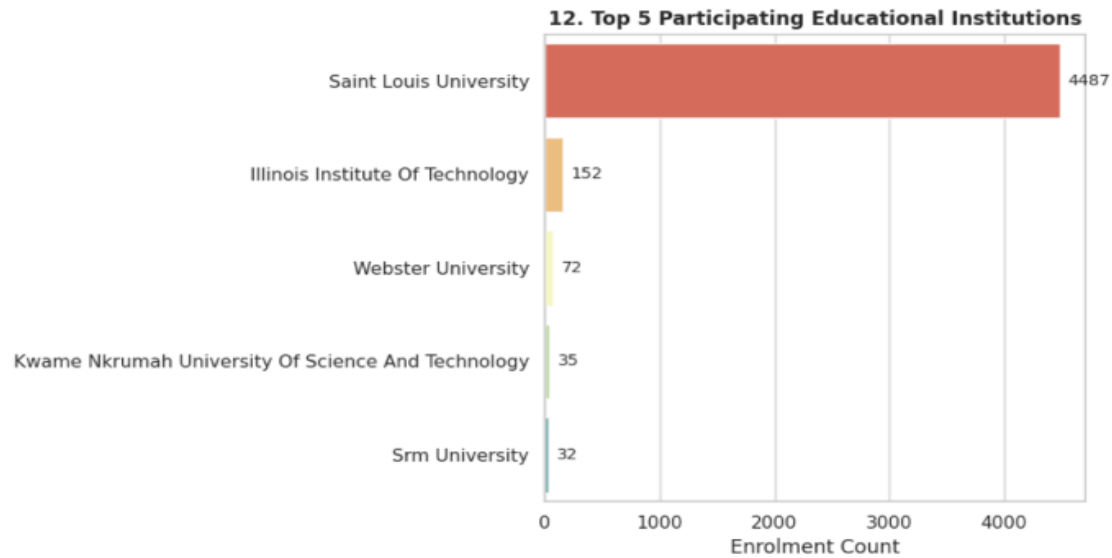


Figure: Top 5 Participating Educational Institutions

Saint Louis University accounts for 4,487 entries, by far the largest single institution in the dataset, dwarfing the next largest (Illinois Institute of Technology, 152). Any institution-level modelling or segmentation should treat this concentration carefully, since it means institution-level patterns are heavily influenced by one school's behaviour.

12. Additional Resources

Resource Type	Format	Description	Document Link / Path
Feature Engineering Notebook	Jupyter Notebook	Complete notebook covering feature engineering,	Click Here: Link
Processed Dataset	CSV File	Final processed dataset used across all Week 2 analyses, including features extracted during	Click Here: Link

		feature engineering.	
Analysis Notebook	Jupyter Notebook	Complete analysis including EDA, trend analysis, and visualizations.	Click Here: Link
Team Charter	Document	Team roles, responsibilities, and working agreement for the project.	Click Here: Link